

DTU Compute
Department of Applied Mathematics and Computer Science

Logical Fallacy Detection using LLMs

Master Thesis

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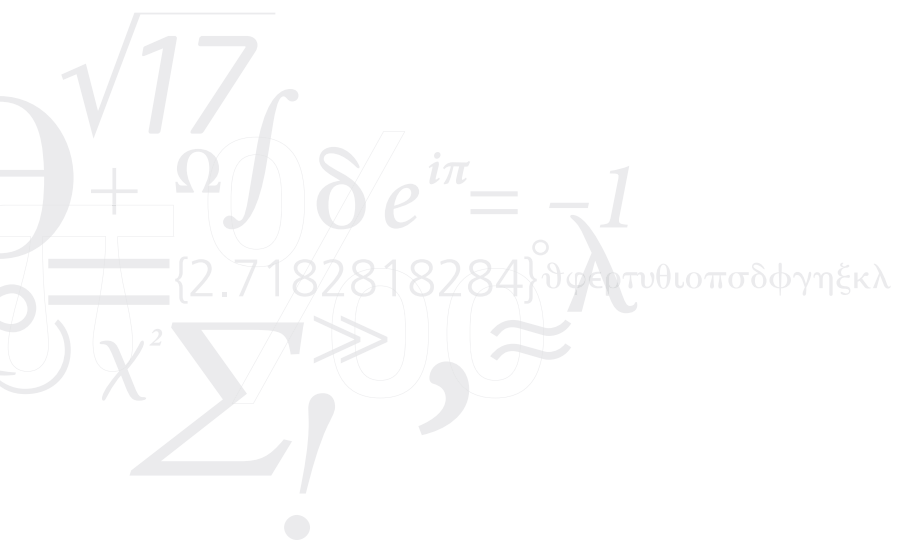


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Abstract

Logical fallacies are a persistent problem in the realm of argumentation, where they tend to contribute to misinformation and flawed reasoning across numerous fields such as media, politics, and online debate. This work investigates the feasibility of using state-of-the-art instruction-tuned large language models (LLMs) to detect logical fallacies. We specifically examine whether LLMs can classify a variety of fallacies in a zero-shot setting, that is, without any fine-tuning on labeled fallacy data. We further explore how prompt design, explicit fallacy definitions, and multi-round classification approaches can influence performance. By employing both Copi's and Aristotle's hierarchical categorization frameworks, the study covers a wide spectrum of broad and subtle fallacy categories to examine each model's capability on the task. Experimental findings show that, while LLMs exhibit moderate success in detecting logical fallacies, they can struggle with more nuanced or similar categories while the prompt sensitivity remains a challenge. Chain-of-thought prompting and hierarchical classification-shifting from broad categories to fine-grained fallacy types- offers a modest increase in classification quality for the strongest models even though overall accuracy dropped. A key additional experiment replaces familiar fallacy names with semantically unrelated labels, revealing that stronger models can compensate for lost semantic clues by relying on contextual cues. These results point to both the promise and current limitations of LLMs in fallacy detection, underscoring the need for improved prompt strategies, external knowledge integration, and more sophisticated reasoning methods to handle the complexity and variety of flawed arguments.

Preface

The present MSc thesis is a partial fulfillment of the requirements for the Master of Science degree in Human-Centered Artificial Intelligence at Technical University of Denmark.

The project accounts for 35 ECTS and is conducted between 4.09.2024 and 04.03.2025. The exact topic along with the objectives of the project are defined in collaboration between the author and the supervisors.

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CHAPTER 1

Introduction

Reasoning is a cognitive process by which humans use prior knowledge to draw inferences, generate explanations, and evaluate statements rationally using logical frameworks [2]. The process is essential for critical thinking, argumentation, and decision-making across various fields, from philosophy to science, public discourse, and artificial intelligence (AI). However, human reasoning is susceptible to systematic flaws that result in logical fallacies—arguments that are potentially convincing but not logically sound.

Logical fallacies are everywhere in human communication and can be observed in day-to-day conversation, mass media, political discourse, and internet forums. Even if they are benign when they result from mere misunderstandings, they are also very effective tools of rhetorical manipulation. The advent of online platforms and social media has, in many respects, amplified the effects of these flawed arguments: misinformation can spread far and wide, and engaging but logically unsound claims can swiftly affect the opinion of the public. Therefore, the need to automate fallacy detection in order to enhance more sensible, evidence-based discussion has reached an all-time high.

During an era of online disinformation, the detection and refutation of logical fallacies are crucial to sound discourse and fact-based decision-making. Fallacy detection, once tested in philosophy and rhetoric, has become a key concern of computational linguistics, artificial intelligence, and information analysis.

This thesis deals with logical fallacy detection using large language models (LLMs). Our approach leverages state-of-the-art, instruction-tuned models in a zero-shot setting. In experimenting with various models, we aim to determine whether modern LLMs have the ability to generalize to identify diverse fallacies without specialized fine-tuning. Here we address the potential of the models in identifying both subtle fallacies as well as broad categories, as outlined by notable classifications such as Copi’s and Aristotle’s frameworks. Furthermore, we try to understand the impact of several prompting strategies on the models’ performance while introducing a multi-round hierarchical classification approach to improve interpretability and accuracy.

The central questions addressed in this thesis are:

- **Zero-Shot Feasibility:** Can large language models reliably detect fallacies even when they have not been trained or fine-tuned on dedicated fallacy corpora?

- **Prompting Strategies:** Does the inclusion of explicit fallacy definitions or changes in prompt design (e.g., prompt wording) significantly impact classification performance?
- **Multi-Round Classification:** Can a hierarchical, chain-of-thought (CoT) approach that moves from coarse- to fine-grained labels improve accuracy and interpretability relative to single-step classification?
- **Semantically Unrelated Labels:** Recognizing that LLMs often rely on semantic cues from familiar label names, we replace standard fallacy labels (e.g., “Ad Hominem,” “Appeal to Emotion”) with abstract placeholders (e.g., “Fallacy_1,” “Fallacy_2”). Comparing performance in this altered label space clarifies how deeply each model depends on memorized label semantics from its training data as opposed to in-context information.

To explore these questions, we conduct a series of experiments on the LOGIC [19] dataset, a benchmark with 13 distinct logical fallacies drawn from a range of sources, including educational materials and online quizzes. Our evaluation includes multiple metrics—accuracy and macro F1 primarily—to account for label imbalance. We also compare model predictions against two baseline approaches: random assignment and distribution-based inference. Ultimately, this work provides insights into the extent to which current LLMs, operating under zero-shot constraints, can address logical fallacy detection, along with the limitations they encounter and the potential strategies for further enhancement.

CHAPTER 2

Background & Related Work

2.1 Fallacies in Argumentation

The systematic study of logical fallacies can be traced back to Aristotle (350 BCE), who, in *Sophistical Refutations* [2], provided one of the earliest known classifications of flawed reasoning patterns. His foundational work laid the groundwork for subsequent inquiries into the nature and structure of fallacious reasoning. Several definitions of argument and fallacy have been introduced; in Figure 2.1 we present the definitions of argument and fallacy as outlined by Copi et al. [6].

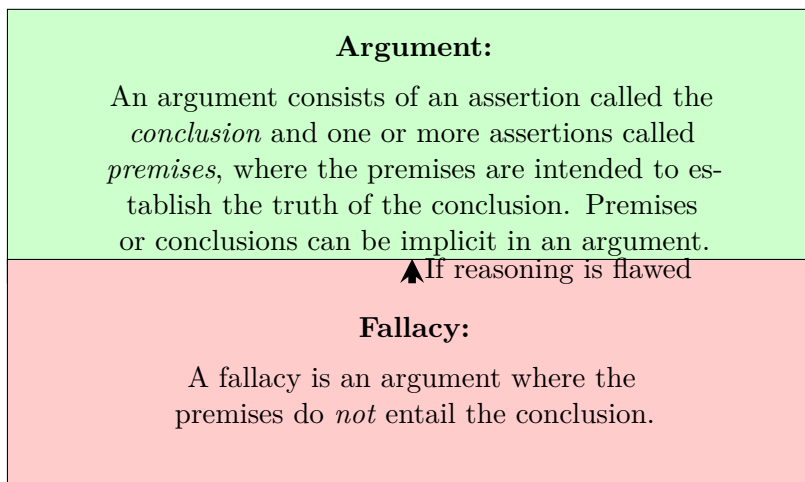


Figure 2.1: Definitions of Argument and Fallacy by Copi et al. [6]

Logical fallacies are not confined to abstract philosophical discourse; they can be found in various spheres of public communication, including:

- **News media** Musi et al. [25], where fallacious arguments are used to shape public perception.

- **Advertising** Danciu et al. [8], where misleading appeals are often employed to persuade consumers.
- **Propaganda and persuasion** Walton [35], particularly in political and ideological discourse.
- **Political rhetoric** Blassnig et al. [3], where fallacies are strategically utilized to influence public opinion.
- **Social media misinformation** Hidayat et al. [17], where logical fallacies contribute to the rapid spread of false narratives.

Given the increasing role of digital platforms in shaping public discourse, the need to effectively identify and mitigate logical fallacies is greater than ever. Advancements in computational argumentation and artificial intelligence have opened new avenues for automating fallacy detection, making the study of fallacies not only a theoretical pursuit but also an essential component of contemporary information analysis.

2.2 Fallacy Detection

The field of logical fallacy detection has seen considerable progress in recent years, driven by the development of novel datasets, innovative methodologies, and advancements in Natural Language Processing (NLP). This section offers a chronological view of the main contributions to the field, highlighting significant milestones in dataset creation and the growing role of large language models (LLMs) in advancing research. By tracing these developments, we aim to provide a comprehensive understanding of how the field has evolved and where it stands today.

In 2017, Habernal et al. [15] were the first to identify the absence of datasets specifically focused on fallacious argumentation. To address this, they developed *Argotario*, a gamified platform aimed at crowdsourcing annotated data for logical fallacies. The platform employed a competitive gameplay model, where participants either created or identified fallacious arguments created by other players. This approach resulted in a dataset encompassing five types of fallacies, including ad hominem and hasty generalization. Argotario functioned both as a data collection tool as well as an educational tool, raising public awareness about logical fallacies. The study demonstrated how gamification could effectively tackle the challenges of data annotation while simultaneously engaging users in meaningful ways.

Da San Martino et al. (2019) [7] introduced the *PROPAGANDA* dataset. Unlike previous approaches that classified entire news articles as propagandistic, they focused on identifying propaganda techniques within text fragments of varying length. This shift allowed for a more interpretable analysis of how propaganda techniques are used in the written news context. The dataset included news articles annotated with 18 distinct propaganda techniques, with some of them being loaded language, name-calling, appeal to fear, and whataboutism, providing a comprehensive framework for

studying these methods in detail. The authors made several key contributions to the field. First, the dataset creation where they curated a corpus of news articles annotated at the fragment level by expert annotators. Second, they proposed two novel tasks: sentence-level classification, where a whole sentence is classified, and fragment-level classification, where a text span is identified as propagandistic alongside the corresponding propaganda technique. Finally, they developed a multi-granularity model that outperformed several strong BERT-based [21] baselines, demonstrating its effectiveness in detecting propaganda at varying levels of granularity. Their approach offered a more precise and explainable framework for analyzing propaganda in news articles.

Vorakitphan et al. (2021) [34] introduced PROTECT (PROpaganda Text dEteC-Tion), propaganda detection and classification tool. PROTECT operates through a two-step pipeline. First, text snippets are identified as propagandistic, and second, these snippets are classified according to the specific propaganda technique employed. To achieve this, the system utilizes transformer-based language models, such as BERT and RoBERTa [21, 24], and integrates semantic and argumentation features to improve classification accuracy. The system was evaluated using the PROPAGANDA dataset developed by Da San Martino et al. (2019) [7], demonstrating state-of-the-art results. Beyond its technical capabilities, PROTECT was designed to be accessible to the public, featuring a user-friendly web interface that enables users to analyze submitted texts. This functionality serves an educational purpose, helping users recognize and understand the manipulative content in media. By providing a practical and accessible tool, this work made a significant contribution to the field, bridging the gap between technical research and public awareness. It highlights the potential of combining advanced NLP techniques with user-centric design to address complex challenges in media analysis and education.

In 2021, Sahai et al. expanded the domain of fallacy detection to online discussions by constructing a dataset from Reddit. The main contribution of this work was the construction of a large, balanced dataset of fallacies. The dataset is created using a two-stage annotation process. First, comments that explicitly mention fallacies were automatically collected from Reddit, and secondly they were manually validated via crowdsourcing. The final dataset consisted of eight common fallacies, including Appeal to Authority, Black-or-White Fallacy, Slippery Slope, and Hasty Generalization. To classify these fallacies, the authors evaluated multiple models, including BERT-based fine-tuned classifiers and the multi-granularity networks introduced by Da San Martino et al. (2019) [7]. In addition, conversational context from the relevant Reddit threads was added which improved classification performance, highlighting the importance of discourse structure in fallacy detection. This work built upon previous research in propaganda detection but unlike prior approaches that focused on detecting fallacies in journalistic articles, Sahai et al. (2021) focused on informal and unstructured online discussions, making their findings particularly relevant for misinformation analysis and online discourse quality assessment, while highlighting the importance of context in fallacy detection.

Jin et al. (2022) [19] introduced the LOGIC dataset, a large-scale benchmark for

detecting 13 types of logical fallacies in natural language text with 2,449 samples. In addition, they propose LOGICCLIMATE, a specialized dataset for detecting fallacies in climate change claims. Their study highlights the difficulty of logical fallacy classification, as existing models exhibit limited performance, achieving only 53.31% micro F1 at best. The dataset includes fallacy types such as Faulty Generalization, Ad Hominem, Appeal to Emotion, False Causality, and Circular Claim, among others. The authors demonstrate that pretrained models (e.g., BERT, RoBERTa, DeBERTa, and Electra) struggle to generalize, especially to out-of-domain fallacies in climate discourse. To improve fallacy detection, they introduce a structure-aware classifier, which distills the argument structure by masking content words and focusing on logical patterns. This approach outperforms previous approaches by 5.46% F1 on LOGIC and 4.51% F1 on LOGICCLIMATE, highlighting the importance of formal logic structures in reasoning tasks. Compared to previous work in propaganda detection (Da San Martino et al., 2019) and fallacy identification in Reddit discussions (Sahai et al., 2021), Jin et al. (2022) contribute a more comprehensive dataset covering multiple fallacy types, making it a significant benchmark for argument quality assessment. Their approach aligns with recent advances in reasoning-aware NLP, providing a foundation for future work in automated argument analysis and misinformation detection.

Alhindi et al. (2022) [1] proposed a unified multitask instruction-based prompting approach for fallacy detection that generalizes across multiple datasets. Unlike previous works, which focused on single-domain fallacy classification [30] [19], their approach integrated five diverse datasets covering 28 fallacy types across different domains (news, dialogue, social media, fact-checking).

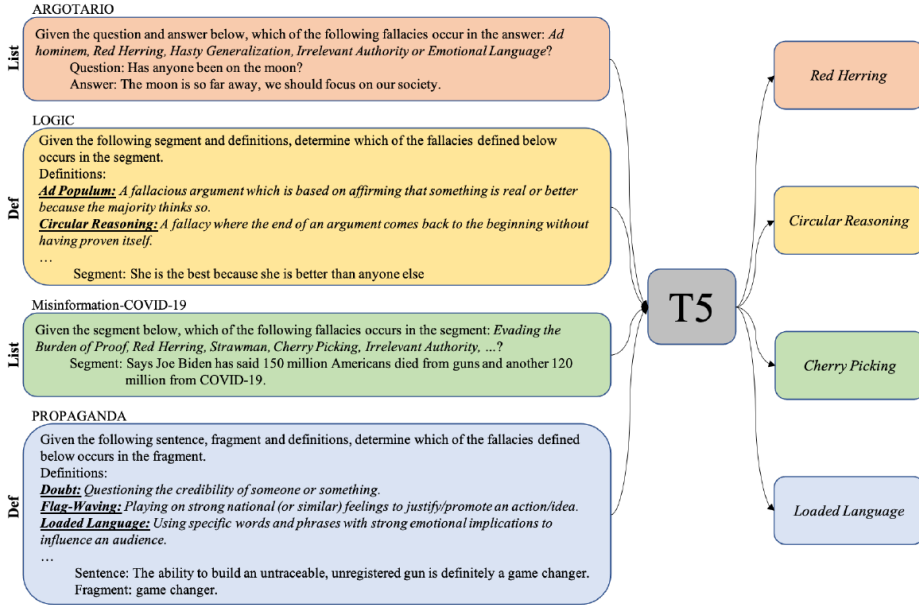


Figure 2.2: Instruction-Based Prompting scheme, source: Alhindi et al. [1]

The authors fine-tuned a T5 transformer model [28] using instruction-based prompting, enabling it to recognize fallacies across various annotation schemes and linguistic structures. Their results demonstrate that multitask prompting significantly outperforms task-specific models, improving macro F1 scores by up to 16% over single-dataset models and achieving better generalization to new domains. Moreover, they analyze the impact of model size, different prompting strategies, and dataset-specific challenges. Furthermore, they conducted a thorough error analysis, revealing the difficulty of the fallacy detection task by comparing an expert annotator's performance to the model's predictions and the dataset labels. The annotator agreed with the dataset's label in 75% of the cases, with the model's predictions in 15% of the cases whereas they provided a different label in 10% of the cases. Compared to earlier approaches that use BERT and RoBERTa-based classifiers for propaganda detection [7] and logical fallacy identification [19], this work provides a more scalable and adaptable model. By leveraging multitask instruction tuning, their system captures cross-domain patterns in fallacious reasoning, making it a valuable contribution to automated argument assessment and misinformation analysis. This study highlights the importance of unified instruction-based prompting for fallacy recognition in real-world settings, emphasizing the need for robust and generalizable models that can handle diverse fallacy types and linguistic structures.

Goffredo et al. (2023) [12] focused on the automatic detection and classification of fallacies in political debates by extending an existing dataset and developing a transformer-based detection model. Their first contribution was the expansion of

the ElecDeb60To16 dataset, a corpus of U.S. presidential debates from 1960 to 2016, by adding the 2020 Trump-Biden debates and incorporating argumentative components (i.e., claims and premises) and the relations between these components (i.e., support and attack). To detect and classify fallacies, the authors proposed MultiFusion BERT, a model architecture combining transformer-based text representations with argumentative features. Their model enhanced fallacy detection by integrating argumentative components and argumentative relations, which improved classification accuracy. Their experimental results showed that MultiFusion BERT outperforms standard transformer-based baselines such as BERT, RoBERTa, and DeBERTa. This work emphasized fallacy detection at the token level, making their approach more applicable to real-world scenarios where fallacious arguments are embedded in unstructured text. By integrating argumentative structure into transformer-based architectures, this study outperformed previous methods. Nevertheless, their work cannot be easily generalized to real world scenarios since the argumentative features are not available in this context.

Finally, while this study was being carried out, a paper trying to answer some of the questions we are trying to answer with this research was released by Pan et al. (2024) [26] exploring the potential for zero-shot fallacy classification with Large Language Models (LLMs), overcoming some of the limitations of standard supervised approaches needing plentiful labeled data and weak performance at generalization. The paper presents some single-round and multi-round prompting techniques to test the reasoning and fallacy detection abilities of state-of-the-art large language models. The authors observe that LLMs can match or even outperform state-of-the-art fine-tuned models in zero-shot scenarios. Single-step prompting where LLMs are directly asked to classify fallacies with or without definitions is competitive, but multi-step prompting—through intermediate reasoning steps such as premise-extraction, definition-generation, and Chain-of-Thought (CoT) reasoning—enhances the performance even for smaller LLMs. One of the main conclusions of the research is that LLMs’ classification capability is a question of the type of dataset: models excel at balanced and easy datasets (e.g., ARGOTARIO, REDDIT), but not at denser fallacies needing more incisive logical examination (e.g., LOGIC). The paper also indicates the tendency of LLMs to confuse similar types of fallacies, revealing the inherent subjectivity of fallacy classification.

Collectively, these studies demonstrate the evolution of fallacy detection, from gamified annotation platforms to sophisticated LLM-based methodologies. Each contribution has addressed unique challenges, from scaling annotation processes to improving model interpretability and reasoning capabilities. These advancements have not only deepened our understanding of logical fallacies but also expanded the applications of NLP in domains such as misinformation detection, argumentation analysis, and critical thinking education.

2.3 Large Language Models (LLMs)

Large language models development has been in the forefront by the evolution of natural language processing (NLP) from statistical methods to deep learning approaches. Traditional methods worked well in basic NLP but had drawbacks of restricted context understanding. Then, though, came the Transformer architecture (Vaswani et al. [33]), which changed the game for NLP by using a self-attention mechanism to enable models to be trained on whole input sequences in parallel. This innovation made training much larger models possible and gave rise to the first generation of LLMs, including OpenAI's GPT [27] and Google's BERT [21].

LLMs have been developed through incremental advances in model scale, training data size, and computational efficiency. Scaling laws research (Kaplan et al. (2020) [20]) showed that model performance improves predictably with increased model size and training data. Subsequent models like GPT-3 (Brown et al., [4]) and LLama (Touvron et al. 2023 [32]) used billion-parameter models, attaining excellent generalization on a wide range of NLP tasks. Furthermore, recent research has proposed new novel architectures such as Mamba (Gu and Dao 2024 [14]) a novel linear-time sequence modeling framework that tackles the computational inefficiencies of standard Transformers while being capable of matching or outperforming Transformers on language datasets while scaling more readily to longer context.

One aspect of the Large Language Models that we are particularly interested in for this work, is their reasoning ability which has been a subject of extensive research. LLMs have been tested both for commonsense causality reasoning [22]. and abstract reasoning [11]. Yet, empirical research [5] has shown that LLMs, even the most recent versions like GPT-4, have inconsistencies in their reasoning.

Commonsense reasoning evaluates the ability of an LLM to apply common sense while making decisions. The work of Kiciman et al. (2023) [22] and Talmor et al. (2020) [31] has evaluated the performance of LLMs on causal understanding tasks. LLMs perform well on formal causal tasks but do poorly on real-world, complex scenarios involving implicit knowledge.

Abstract reasoning, as studied by Gendron et al. (2023) [11], which consists of finding and applying a general pattern from few data, argued that while models excel at syntactic pattern recognition, they often fail at true abstract generalization.

Recently, Bubeck et al. [5], despite their amazing performance in several tasks, has identified essential limitations in large language models (LLMs), specifically their consistency in reasoning tasks. Even the most advanced models (GPT-4) tend to derive contradictory conclusions from differences in prompts, suggesting that their reasoning is shaped more by the wording of inputs than by a deep understanding of logical principles.

Building on these findings, we find it essential to examine the reasoning capabilities of LLMs in detecting logical fallacies, a task that requires a solid understanding of logical principles and the ability to detect flawed patterns of reasoning. Through an evaluation of LLMs' performance in fallacy detection tasks, we aim to understand their reasoning abilities and identify potential areas of improvement.

2.4 Chain of Thought

Chain-of-Thought (CoT) prompting has greatly enhanced the reasoning ability of large language models (LLMs) by allowing them to decompose challenging problems into a series of intermediate steps. Conventional LLMs are usually poor at multi-step reasoning tasks, where they give straightforward but sometimes wrong responses because of the absence of systematic cognitive processes. CoT prompting overcomes this restriction by promoting the generation of explicit intermediate reasoning steps in models, thereby mimicking human-like logical reasoning.

The concept of Chain-of-Thought (CoT) prompting was originally defined by Wei et al. (2022) [36], where they demonstrated that breaking complex issues into consecutive reasoning steps significantly improved the performance of large language models (LLMs) on multi-step reasoning tasks. Their research demonstrated that if LLMs were prompted to provide intermediate reasoning steps, they were better on math word problems and logical reasoning tasks. The innovation diverged from the classical "end-to-end" responding mode of LLMs, where generally immediate answers were produced by models without actually reasoning out the problem.

Follow-on research extended optimization of CoT prompting on varied reasoning tasks. Kojima et al. (2022) [23] in their effort to address the significant drawback in Wei et al.'s method which was the requirement for manually created exemplars, showed that when LLMs were prompted to "think step by step," generated sizeable increases in logical consistency. The authors of this paper demonstrated that reasoning using CoT-type reasoning is effective on reasoning-trained models but can also automatically occur on very large LLMs.

Zhou et al. (2023) [38] built upon the research of CoT prompting and formulated Least-to-Most (LtM) prompting. LtM tries to address CoT's problems in handling easier and harder tasks. Unlike standard CoT, which solves a problem step by step, LtM first breaks down a difficult task into easier, smaller components and solves them in turn, making use of prior responses along the way. This method substantially improves compositional generalization and multi-step reasoning performance compared to conventional CoT prompting. Through hierarchical structuring of problem-solving, LtM improves the flexibility and development of reasoning in LLMs, bringing AI closer to human thinking abilities.

The CoT prompting studies have shown that it is a highly effective technique for enhancing the reasoning capability of LLMs to enable them to resolve difficult multi-step problems more accurately and consistently. By breaking down problems into a series of intermediate steps, CoT prompting mimics human-like logical reasoning, providing a systematic problem-solving approach that can significantly improve the performance of LLMs on an extremely wide range of reasoning tasks. Therefore, we decided to use this technique in our work to explore how different chain-of-thought techniques could enhance the capabilities of LLMs in detecting logical fallacies.

CHAPTER 3

Method

In this chapter we describe the methodology used to evaluate the performance of the models in the task of logical fallacy detection. We begin by describing the dataset used for the experiments, followed by the models and the evaluation metrics used. We then describe the experimental setup, including the zero-shot, the categorization of the fallacies included in our dataset of choice, the chain-of-thought prompting and the multi-round chain of thought approach used to evaluate the models' performance. Finally, we perform one more experiment with semantically unrelated labels to evaluate the models' performance in the absence of familiar fallacy names.

3.1 Dataset

We use the **LOGIC** dataset for this study. Jin et al. (2022) [19] introduced the dataset, and it includes 2,449 examples of 13 logical fallacies. The dataset is gathered from various sources, including educational materials and online quizzes. The variety of different logical fallacies included renders it highly useful for evaluating the overall performance of models in the fallacy detection task.

Logical Fallacy	Examples
Faulty Generalization (18.01%)	"I met a tall man who loved to eat cheese. Now I believe that all tall people like cheese." "Sometimes flu vaccines don't work; therefore vaccines are useless ."
Ad Hominem (12.33%)	"What can our new math teacher know? Have you seen how fat she is?" "I cannot listen to anyone who does not share my social and political values."
Ad Populum (9.47%)	"Everyone should like coffee: 95% of teachers do!" "Killing thousands of people as a result of drug war campaign is not a crime to humanity because millions of Filipino support it."
False Causality (8.82%)	" Every time I wash my car, it rains. Me washing my car has a definite effect on the weather." "Every severe recession follows a Republican Presidency; therefore Republicans are the cause of recessions."
Circular Claim (6.98%)	"J.K. Rowling is a wonderful writer because she writes so well ." "She is the best candidate for president because she is better than the other candidates!"
Appeal to Emotion (6.82%)	"It is an outrage that the school wants to remove the vending machines. This is taking our freedom away! " "Vaccines are so unnatural; it's disgusting that people are willing to put something like that in their body."
Fallacy of Relevance (6.61%)	" Why are you worried about poverty? Look how many children we abort every day." "Why should we be worrying about how the government treats Native people, when people in our city can't get a job"
Deductive Fallacy (6.21%)	" It is possible to fake the moon landing through special effects. Therefore , the moon landing was a fake using special effects." "Guns are like hammers—they're both tools with metal parts that could be used to kill someone. And yet it would be ridiculous to restrict the purchase of hammers, so restrictions on purchasing guns are equally ridiculous ."
Intentional Fallacy (5.84%)	" No one has ever been able to prove that extraterrestrials exist, so they must not be real." "It's common sense that if you smack your children, they will stop the bad behavior. So don't tell me not to hit my kids."
Fallacy of Extension (5.76%)	"Their support of the discussion of sexual orientation issues is dangerous: they advocate for the exposure of children to sexually explicit materials, which is wrong ." "They say we should cut back the defense budget. Their position is that they want to leave our nation completely defenseless!"
False Dilemma (5.76%)	"You're either for the war or against the troops." "I don't want to give up my car, so I don't think I can support fighting climate change."
Fallacy of Credibility (5.39%)	"My professor , who has a Ph.D. in Astronomy , once told me that ghosts are real. Therefore, ghosts are real ." "My minister says the Covid vaccine will cause genetic mutations. He has a college degree , and is a holy man, so he must be right ."
Equivocation (2.00%)	"I don't see how you can say you're an ethical person . It's so hard to get you to do anything; your work ethic is so bad" "It is immoral to kill an innocent human being. Fetuses are innocent human beings. Therefore, it is immoral to kill fetuses."

Figure 3.1: Examples of the 13 logical fallacy types and their distribution in the LOGIC dataset. Examples with neutral impact that are highlighted with black whereas the ones with potentially negative impact are highlighted with red. source: Jin et al. (2022) [19]

Furthermore, The LOGIC dataset is appropriate for our work as it provides labeled text segments of fallacious reasoning and it's distinct from other datasets that address specific argument flaws, e.g., ad hominem fallacies. LOGIC involves many various types of fallacies, so it is a general benchmark for the fallacy detection task. Additionally, the past experiments on this dataset showed that even state-of-the-art fine-tuned language models find it hard to perform well. This fact makes LOGIC dataset the most challenging one in the logical fallacy detection task and appropriate for our task to explore how well LLMs can perform and whether we can boost their performance by using different techniques.

3.2 Models

For our experiments, we selected Falcon-Mamba-7B-Instruct, Meta-Llama-3.1-8B-Instruct, and Mistral-7B-Instruct-v0.3 as our primary models. Since our work involves long prompts—including fallacy names, definitions, and multi-round prompting—we needed models with strong long-context retention and instruction-following capabilities. These models were chosen for their ability to process and reason over extended input sequences while maintaining efficiency and logical coherence. Their instruction-tuning capabilities and proficiency in zero-shot/few-shot reasoning make them particularly well-suited for fallacy detection task.

Falcon-Mamba-7B-Instruct represents a breakthrough in non-Transformer architectures, employing state-space modeling (SSM) instead of self-attention to enable scalable, memory-efficient long-context processing (Zuo et al., 2024) [39]. Unlike traditional LLMs, which face quadratic complexity in attention computation, Falcon-Mamba-7B uses a structured recurrence mechanism to maintain constant memory usage regardless of sequence length. This makes it highly effective for document-level reasoning and real-time applications. Additionally, it has been trained on a massive 5.8 trillion token dataset, incorporating instruction tuning, multilingual data, and domain-specific corpora, further enhancing its accuracy in complex tasks.

Meta-Llama-3.1-8B-Instruct, part of the LLaMA (Large Language Model Meta AI) family, extends Transformer-based architectures by integrating Grouped Query Attention (GQA) and Rotary Positional Embeddings (RoPE), leading to improved efficiency in reasoning and long-context comprehension (Meta AI, 2024) [13]. Trained on 15 trillion tokens, this model is fine-tuned for multilingual applications, logical inference, and structured decision-making tasks. Empirical benchmarks show that Meta-Llama-3.1-8B-Instruct outperforms earlier LLaMA models in several tasks including math reasoning and multilingual benchmarks.

Mistral-7B-Instruct-v0.3 is a highly optimized Transformer model designed to balance reasoning accuracy with inference efficiency, making it a strong alternative to larger LLMs (Jiang et al., 2023) [18]. It incorporates Sliding Window Attention (SWA) and Grouped Query Attention (GQA) to preserve context over long sequences while maintaining fast inference speeds, which is particularly beneficial for multi-step logical reasoning. Despite its relatively compact 7-billion-parameter size, Mistral-7B-Instruct delivers competitive results in code generation, commonsense reasoning, and structured question-answering, outperforming larger models while using significantly fewer computational resources.

The selection of Falcon-Mamba-7B-Instruct, Meta-Llama-3.1-8B-Instruct, and Mistral-7B-Instruct-v0.3 was driven by their superior long-context reasoning capabilities, efficient processing mechanisms, and strong instruction-following performance. Each model brings distinct architectural strengths that contribute to handling the complexity of logical fallacy detection, particularly when working with extended prompts and multi-step reasoning tasks.

3.3 Zero-Shot

At first, we conducted a set of zero-shot experiments with the above-stated models. Zero-shot experiments test the capacity of a model to carry out a task without having been specifically trained on that task, thus proving to be an effective method for assessing whether these models are capable of performing this task without any fine-tuning. To test their performance and consistency, based on previous studies suggesting that the wording of a prompt can significantly affect the model's response, we designed and executed two different prompts for each of the models asking for the same task but with some minor differences in wording and on the specific instructions. The prompts, as shown in Table 3.1, were specifically designed to test the models' capacity for classifying fallacious arguments regardless of explicit supervision or exposure to labeled training data.

Furthermore, to examine the possible effect of explicit definitional instruction on the performance of the models, we included two more variations for each prompt. One variation included full definitions of the logical fallacies, with explicit explanations and contextual information about the nature and structure of each fallacy. In contrast, the second variation explicitly excluded these definitions, leaving the models to depend on their prior knowledge on the topic and reasoning capabilities alone to detect instances of fallacious argumentation.

Prompt 1	Prompt 2
You will be provided with a text segment, delimited by ##### characters, and your task is to classify it into one of the fallacy types listed below. Provide your output in JSON format with the key: 'detected_fallacy'. The value should be the name of the detected fallacy, chosen from the list below. Do not include any explanation or commentary. The ##### characters are only delimiters for the segment. Do not include them in your classification or output. The possible fallacy types are: 1. Appeal To Emotion: attempting to arouse non-rational sentiments within the intended audience in order to persuade. 2. False Causality: occurs when someone mistakenly assumes that because one event follows another, the first event caused the second, without sufficient evidence for a causal link. 3. Ad Hominem: The opponent attacks a person instead of arguing against the claims that the person has put forward. 4. Faulty Generalization: A generalization is drawn from a sample which is too small, it is not representative of the population or it is not applicable to the situation if all the variables are taken into account. 5. False Dilemma: Presenting two alternative options as the only possibilities, when in fact more possibilities exist. As an the extreme case, tell the audience exactly what actions to take, eliminating any other possible choices. 6. Fallacy of Relevance: The argument supporting the claim diverges the attention to issues which are irrelevant for the claim at hand. 7. Fallacy of Credibility: happens when someone argues that a claim is true simply because an authority or expert believes it, even if that authority is not a reliable or relevant source on the topic. 8. Ad Populum: A fallacious argument which is based on affirming that something is real or better because the majority thinks so. 9. Circular Reasoning: A fallacy where the end of an argument comes back to the beginning without having proven itself. 10. Fallacy of Extension: An argument that attacks an exaggerated or caricatured version of your opponent's position. 11. Equivocation: An argument which uses a key term or phrase in an ambiguous way, with one meaning in one portion of the argument and then another meaning in another portion of the argument. 12. Fallacy of Logic: An error in the logical structure of an argument. 13. Intentional Fallacy: Some intentional (sometimes subconscious) action/choice to incorrectly support an argument. The segment to be classified is the following: #####segment##### Provide your output in strict JSON format with the key 'detected_fallacy'. For example: { "detected_fallacy": "name of fallacy" }	You are a knowledgeable expert in analysing fallacies in discourses. Please ensure that your responses are socially unbiased in nature. Your response should not be lengthy. Answer the last question. Based on the following definitions of fallacies: 1. Appeal To Emotion: attempting to arouse non-rational sentiments within the intended audience in order to persuade. 2. False Causality: occurs when someone mistakenly assumes that because one event follows another, the first event caused the second, without sufficient evidence for a causal link. 3. Ad Hominem: The opponent attacks a person instead of arguing against the claims that the person has put forward. 4. Faulty Generalization: A generalization is drawn from a sample which is too small, it is not representative of the population or it is not applicable to the situation if all the variables are taken into account. 5. False Dilemma: Presenting two alternative options as the only possibilities, when in fact more possibilities exist. As an the extreme case, tell the audience exactly what actions to take, eliminating any other possible choices. 6. Fallacy of Relevance: The argument supporting the claim diverges the attention to issues which are irrelevant for the claim at hand. 7. Fallacy of Credibility: happens when someone argues that a claim is true simply because an authority or expert believes it, even if that authority is not a reliable or relevant source on the topic. 8. Ad Populum: A fallacious argument which is based on affirming that something is real or better because the majority thinks so. 9. Circular Reasoning: A fallacy where the end of an argument comes back to the beginning without having proven itself. 10. Fallacy of Extension: An argument that attacks an exaggerated or caricatured version of your opponent's position. 11. Equivocation: An argument which uses a key term or phrase in an ambiguous way, with one meaning in one portion of the argument and then another meaning in another portion of the argument. 12. Fallacy of Logic: An error in the logical structure of an argument. 13. Intentional Fallacy: Some intentional (sometimes subconscious) action/choice to incorrectly support an argument. Given a segment of discourse below, determine which of the fallacies defined above is present in the argument? Segment: segment Respond strictly in JSON format. For example: { "detected_fallacy": "name of fallacy" } Do not include any explanation or additional commentary.

Table 3.1: Comparison of our two Fallacy Detection Prompts (including definitions). With red we highlight the main differences between the two prompts.

The design of the experiment was conceived to consider two main research questions. First, we aimed to find out if the models are able to recognize logical fallacies correctly without being provided with explicit definitions. This part of the experiment enabled us to assess the extent to which pre-trained language models have an internalized appreciation of both logical reasoning and fallacious argumentation derived from their previous training data alone. Second, we wished to measure the degree to which the addition of formal definitions affected the performance of the models. Therefore, we tested whether providing explicit definitions would have a measurable boost in accuracy and consistency in the classification by the models.

This methodological approach enabled us to develop a clearer understanding of not only the potential but also the limitations of zero-shot learning when it comes to logical fallacy detection.

3.4 From fine-grained to coarse grained classes

By structuring the fallacies in a hierarchical manner, we aimed to explore how well the models could generalize their reasoning across different levels of categorization. Specifically, we sought to determine whether the models exhibited stronger classification abilities at the coarse-grained level (recognizing broad fallacy types) or the fine-grained level (distinguishing between individual fallacies within each category). This approach allowed us to assess the extent to which language models understand logical fallacy classification beyond simple keyword recognition, providing deeper insights into their reasoning capabilities.

3.4.1 Coarse-grained classes by Copi

Copi's [6] classification system of fallacies of reasoning comes from formal and informal logic, an ordered method toward evaluating faulty reasoning. Copi's classification revolves around argument soundness and argument validity, placing it as an essential cornerstone for modern educational materials on logic and critical thinking. Copi classifies fallacies of reasoning under four general categories, namely, Fallacies of Relevance, Fallacies of Defective Induction, Fallacies of Presumption, and Fallacies of Ambiguity. The mappings between the fine-grained LOGIC classes and these categories can be found in Figure 3.2.

Fallacies of Relevance involve premises that bear no relevance to an argument's conclusion, even if these premises sound plausible. *Ad hominem*, an attack against the speaker, not the claim, *ad populum*, an argument that makes an appeal to people, not facts, and *appeal to emotion*, an argument playing on emotion, not reason, fall under this classification. Relevance fallacies, such as intentional fallacy, an argument distorting an argument's meaning so as to change an argument's emphasis, fall under this classification too.

Fallacies of Defective Induction occur when an argument makes an unwarranted conclusion from incomplete or unreliable evidence. These fallacies involve *false causal*-

ity, assuming causation without proof; *false dilemma*, posing an incomplete list of choices when there are many, many more; and *faulty generalization*, drawing general conclusions from an incomplete, unrepresentative sample. Others involve the *fallacy of credibility*, relying upon unverified authority, and *deductive fallacy*, applying the rule of reasoning improperly.

Fallacies of Presumption involve argument presupposing that what it seeks to confirm, including *circular reasoning*, where an argument's conclusion is contained within the premise(s).

Lastly, Fallacies of Ambiguity play upon linguistic uncertainty, utilizing imprecise, deceptive language, including *equivocation*, shifting the terms' meaning in an argument.

Copi's classification provides an orderly way of analyzing reasoning faults, making it an invaluable tool for assessing arguments.

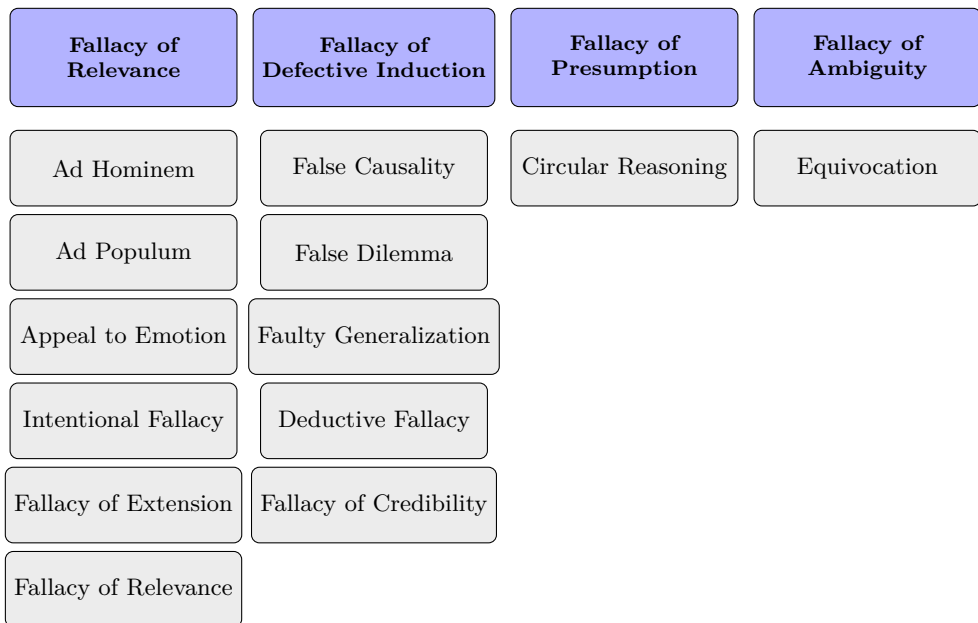


Figure 3.2: Copi's Categorization of Logical Fallacies

3.4.2 Coarse-grained classes by Aristotle

The following classification of logical fallacies has its roots in Aristotle's [37] three modes of persuasion—Ethos (credibility), Pathos (emotion), and Logos (logic)—as described in his work *Rhetoric* [29]. These modes represent the fundamental ways in which arguments persuade an audience, and they serve as the basis for Aristotle's categorization of rhetoric schemes. We use these categories to classify the fallacies

of the LOGIC dataset based on the work of Helwe et al. (2024) [16] and the exact mapping between the fine-grained logic classes to these categories can be found in Figure 3.3.

The Fallacy of Credibility (Ethos fallacies) occurs when an argument cites an untrustworthy source or authority over sound reasoning. *Ad hominem*, whereby the speaker attacks one’s character, not one’s argument, *ad populum*, whereby one cites popular opinion as proof of truth, and the *fallacy of credibility*, whereby an argument rests upon nothing other than an authority figure’s approval, even if the authority figure has no expertise, fall under this category. These fallacies play upon apparent credibility, not verifiable facts, so are very influential in political and social contexts.

The Fallacy of Logic (Logos-based fallacies) entails reasoning fallacies. Some fallacies that fall under this category are *false causality*, wherein causation gets mixed up with correlation; *false dilemma*, wherein only two out of several possible choices is shown; and *faulty generalization*, wherein a conclusion is drawn from an incomplete, unbalanced sample. Other fallacies here are *circular reasoning*, wherein an argument takes an argument as one of its premises, and *equivocation*, wherein one utilizes ambiguous terms.

Lastly, the Fallacy of Emotion (Pathos-based fallacies) utilizes emotional appeal to persuade, not reason. This category involve *appeal to emotion*, wherein one utilizes emotion over facts; *intentional fallacy*, wherein one distorts one’s intent as proof of one’s accuracy; and *fallacy of extension*, wherein one distorts an opponent’s argument so it is easy to demolish.

By categorizing fallacies under these categories of rhetoric, this classification has an orderly means of analyzing faulty reasoning, as well as an excellent tool by which one may analyze how large language models discriminate various logical fallacies.

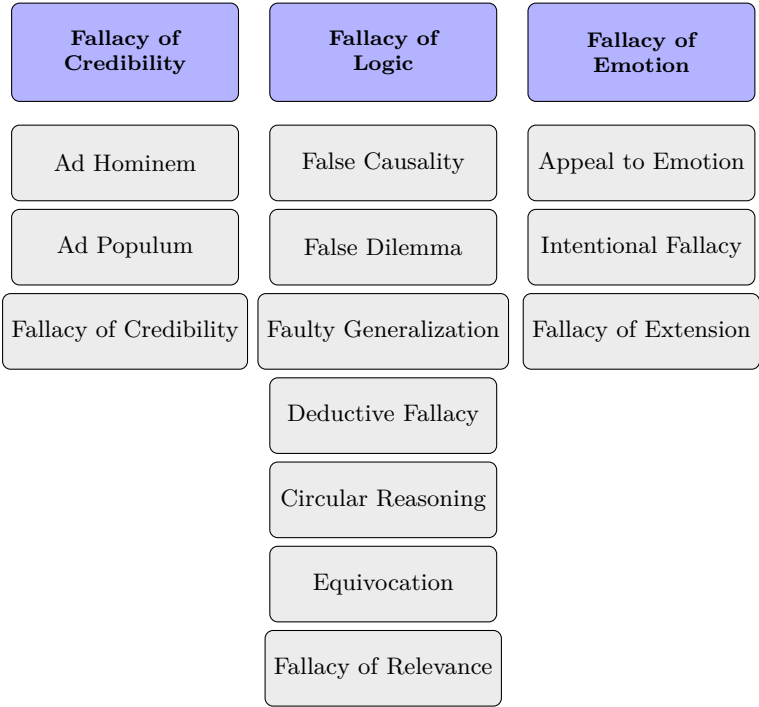


Figure 3.3: Aristotle’s Categorization of Logical Fallacies

3.5 Chain of thought

As described in Chapter 2, Chain-of-Thought (CoT) prompting has significantly advanced the reasoning capability of large language models (LLMs) on wide-ranging tasks. With its success in logical inference and structured reasoning, we decided to use CoT prompting to the task of fallacy detection.

Specifically, we utilized the method developed by Kojima et al. (2023) [23], which demonstrated that even a simple instruction—“Let’s think step by step”—could considerably improve model performance on zero-shot reasoning tasks. By adding this instruction to our prompts, we wanted to test whether asking the model to break down its reasoning process would enhance its ability to detect logical fallacies. This approach allowed us to explore whether LLMs could improve their reasoning in fallacy detection without requiring additional training or fine-tuning.

By comparing the model’s responses with and without CoT prompts, we aimed to determine how much step-by-step reasoning contributes to more reliable fallacy detection. The findings could provide insight into how reasoning-based prompting techniques can be leveraged to enhance LLM performance in the fallacy detection task.

3.6 Multi-round chain of thought, from coarse-grained to fine-grained fallacy types

In this section, we introduce a multi-round chain of thought approach inspired by the work and promising results of Pan et al. (2024) [26] to test how accurately the model performs in logical fallacy classification. Based on Least-To-Most prompting introduced by Zhou et al. [38] we decided to use the categorizations described above to introduce a multi-round chain of thought experiment. This approach is designed to test the model’s ability to classify logical fallacies in a step-by-step manner, starting from coarse-grained categories and moving towards fine-grained categories. This method allows us to evaluate whether the model’s performance would increase if we start from an easier task and move towards a more difficult one.

The experiment begins by having the model label a given text segment as falling into one of the coarse-grained categories of fallacies listed in the previous sections. Following this, the model is presented with the fine-grained fallacy categories that fall within the selected coarse-grained category and it is asked to select the most appropriate fine-grained fallacy category. This step is crucial as it tests if the model is able to make more specific classifications by using the taxonomy that has been presented to it.

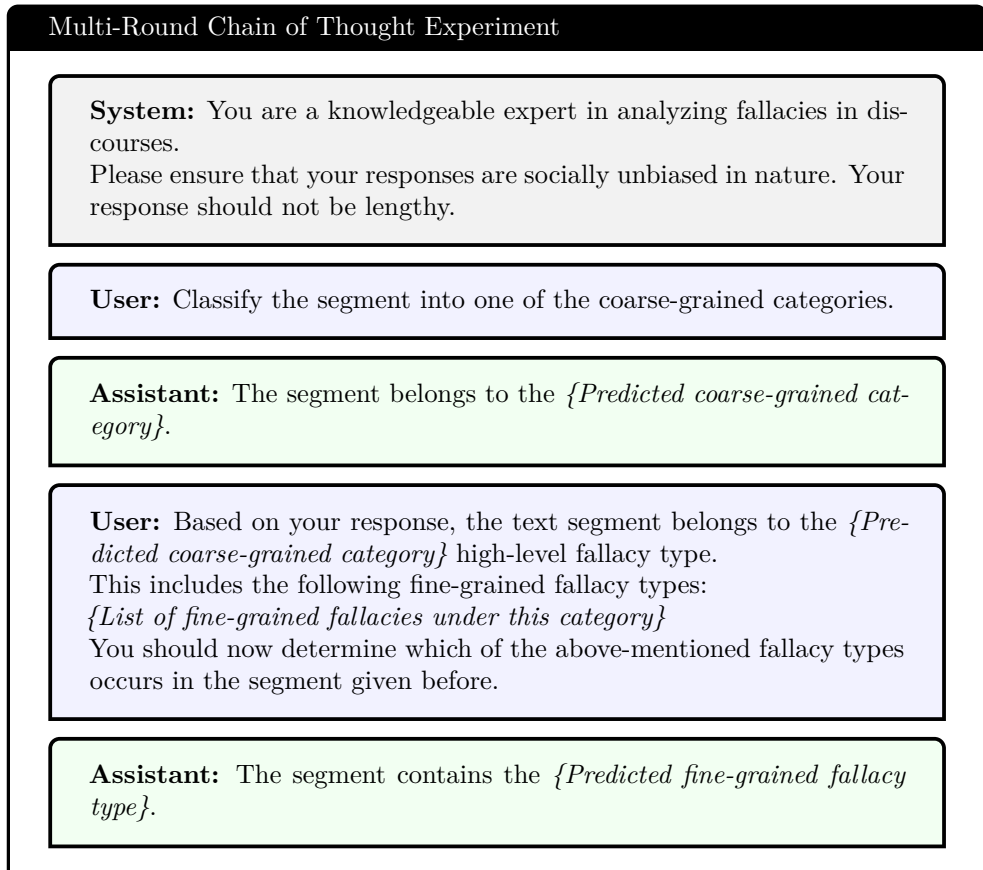


Figure 3.4: Multi-Round Chain of Thought Prompt

By structuring classification in a hierarchical multi-round style, we attempt to determine if iterative prompting—having the model prompted successively through broader to more refined classes—is more effective than having it classify in a single attempt. This approach allows us to examine the model’s reasoning capacity through a step-by-step thinking process, from a broad categorization to a narrower one. Therefore, it offers valuable insight into the model’s reasoning process and its capacity for fine distinction in logical fallacy detection.

3.7 Semantically Unrelated Labels

To further investigate the extent to which the models rely on their internal representations from training data versus newly provided in-context information, we designed

an additional experiment. In this setup, each model was presented with a structured list of fallacies (e.g., “Fallacy_1: definition,” “Fallacy_2: definition,” etc.) before classifying a given text segment into one of these fallacies. By explicitly supplying these definitions within the prompt, we aimed to understand whether and how each model integrates newly introduced information into its reasoning process. Comparing performance under these conditions with earlier results offered deeper insights into the models’ capability for generalization, as well as the relative contributions of in-context reasoning versus pretrained knowledge in fallacy detection tasks.

CHAPTER 4

Experiments

This chapter presents a comprehensive analysis of our approach to logical fallacy detection with large language models (LLMs). We first outline the experimental setup, such as the characteristics of the dataset, evaluation metrics, and baseline approaches adopted. Then we cover the model performance at the fine-grained and coarse-grained levels, namely the distinction between macro F1 and accuracy, the impact of presenting explicit fallacy definitions, and the influence of prompt design. We also discuss the results of the chain-of-thought (CoT) reasoning and hierarchical classification methods, which aim to improve interpretability and accuracy by breaking down the classification task into multiple steps. Finally, we analyze the models' performance when the familiar fallacy labels are replaced with abstract placeholders, shedding light on the extent to which the models rely on memorized label semantics versus in-context learning.

4.1 Experimental Setup

This section provides an overview of the experimental setup, including the dataset, model selection, and evaluation metrics.

4.1.1 Dataset

Due to resource constraints, we focused our analyses exclusively on the test subset of the LOGIC dataset (Jin et al. [19]), which is comprised of 300 text segments labeled with 13 different logical fallacies. All experiments were conducted on DTU's high-performance computing (HPC) cluster [9], using Tesla A100 GPUs equipped with 40 GB of VRAM. Each model required approximately 15 GB of VRAM to run, and generating each of the tables presented in the following sections took around six hours. Consequently, running the same set of experiments on the entire dataset (around 3000 samples) would require 60 continuous hours per table, translating to roughly 600 GPU hours (approximately 25 days) for the 10 tables alone—excluding any preliminary test runs. Such an allocation, using half the VRAM of a single GPU for nearly a month, would be infeasible given the shared job queue system at DTU. The test subset, which is representative of the full dataset, therefore serves as a sufficient resource for evaluating the performance of our models.

The distribution of the original dataset’s fine grained classes on this subset can be found in Figure 4.1. The most common fallacies are *Faulty Generalization*, *Ad Hominem* and *Ad Populum* whereas the least common ones are *Equivocation* and *False Dilemma*.



Figure 4.1: LOGIC Dataset Distribution (Test Subset)

In Figure 4.2, the distribution of the coarse-grained classes by Copi in the LOGIC dataset’s test subset is shown. The dataset is highly imbalanced, with the most common classes being *Fallacy of Relevance* and *Fallacy of Defective Induction*.

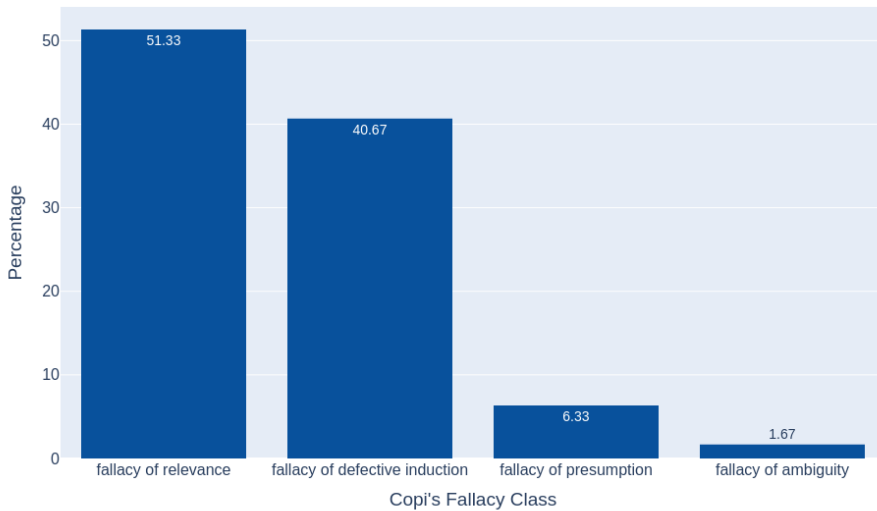


Figure 4.2: LOGIC Dataset to coarse-grained classes by Copi Distribution (Test Subset)

Lastly, the distribution of the coarse-grained classes by Aristotle in the LOGIC dataset's test subset can be found in Figure 4.3. The dataset is highly imbalanced in this case too, with the most common fallacy type being *Fallacy of Logic* with about half of the dataset belonging to this class.

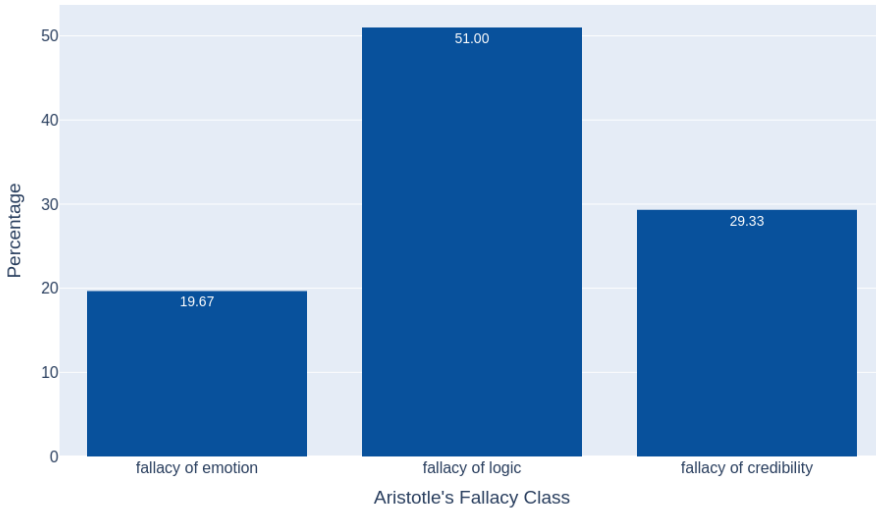


Figure 4.3: LOGIC Dataset to coarse-grained classes by Aristotle Distribution (Test Subset)

From the figures above it is clear that the dataset is highly imbalanced, with some classes being more common than others. This is important to keep in mind when evaluating the performance of the models, as the macro F1 score provides a clearer insight into how each model handles minority classes.

4.1.2 Model Selection and Baselines

We evaluate three instruction-tuned large language models: Falcon-Mamba-7B-Instruct, Meta-Llama-3.1-8B-Instruct, and Mistral-7B-Instruct-v0.3 using the API offered by huggingface [10]. Each model is tested under zero-shot conditions without any additional fine-tuning on fallacy-related corpora. We compare these models against two baselines: a random baseline that assigns each instance a label uniformly at random among the 13 possible classes, and a distribution baseline that predicts labels proportionally to their frequencies in the dataset. Both baselines serve as references for how well the models perform compared to random guessing and a simple distribution-based approach.

4.1.3 Evaluation Metrics

We focus on two primary metrics:

1. **Accuracy:** the fraction of correctly classified instances among all test samples.

2. **Macro F1**: the average of the F1 scores across all categories, giving equal emphasis to minority classes.

Due to the dataset’s imbalance, macro F1 is especially informative in revealing whether a model disproportionately favors common fallacies. We also examine the effect of Definitions (providing or omitting a brief explanation of each fallacy) and compare two different Prompt variants (Prompt 1 vs. Prompt 2). Both prompts request fallacy classification in JSON format, but they differ in wording and instructions. We systematically vary these conditions to evaluate how they influence model performance.

Furthermore, we report two more metrics in our tables. The failed cases and the unknown labels. The failed cases correspond to the percentage of examples that we failed to parse the model’s response which was asked to be in a json format. The unknown labels correspond to the percentage of examples that the model classified using other labels not included in the dataset.

4.2 Zero-shot

As described in the previous section we first conducted zero-shot experiments to evaluate the performance of the models on the LOGIC dataset. We used two different prompts denoted as prompt 1 and prompt 2. Furthermore, we ran the experiments both with and without the definitions of the fallacy classes. The results are presented in the following tables.

4.2.1 Fine Grained Logic Classes

model	def	prompt	accuracy (%)	F1-score	failed (%)	unknown (%)
Random-Baseline	N/A	N/A	7.966	7.160	0.000	0.000
Distribution-Baseline	N/A	N/A	10.466	0.074	0.000	0.000
falcon-mamba-7b-instruct	False	1	21.000	0.081	2.333	7.667
falcon-mamba-7b-instruct	True	1	20.333	0.086	2.667	11.667
Meta-Llama-3.1-8B-Instruct	False	1	40.333	0.211	0.000	2.333
Meta-Llama-3.1-8B-Instruct	True	1	38.333	0.208	0.000	1.667
Mistral-7B-Instruct-v0.3	False	1	39.333	0.153	0.000	6.333
Mistral-7B-Instruct-v0.3	True	1	34.667	0.210	0.000	7.333
falcon-mamba-7b-instruct	False	2	24.667	0.095	6.333	9.000
falcon-mamba-7b-instruct	True	2	22.667	0.082	6.000	13.000
Meta-Llama-3.1-8B-Instruct	False	2	40.000	0.216	0.000	2.333
Meta-Llama-3.1-8B-Instruct	True	2	41.667	0.237	0.000	1.333
Mistral-7B-Instruct-v0.3	False	2	39.333	0.158	0.000	4.667
Mistral-7B-Instruct-v0.3	True	2	33.000	0.189	0.000	8.333

Table 4.1: Fine-grained Classes Results. In this table the zero-shot results using the original LOGIC dataset’s labels are presented.

The findings presented in Table 4.1 indicate considerable variation in the performance of the different models, while showing that all the models outperformed the random baselines. Meta-Llama-3.1-8B-Instruct consistently outperformed its counterparts by recording the highest accuracy of 41.667% and macro F1 score of 0.237 with prompt 2 and the included definitions. On the other hand, Falcon-Mamba-7B-Instruct presented the most difficulties, with the lowest accuracy and F1 score under all the conditions tested, as well as with higher failure rates and more unknown labels. Mistral-7B-Instruct-v0.3 presented a more unstable performance, as its F1 score varied in relation to the prompt and with or without definitions. Worth mentioning is the fact that both Meta-Llama-3.1-8B-Instruct and Mistral-7B-Instruct-v0.3 had a 0% failure rate, whereas Falcon-Mamba-7B-Instruct frequently struggled to give well-formatted answers, especially for prompt 2.

A trend that can be seen from the results is that prompt 2 generally led to better performance than prompt 1 even though there are some exceptions, suggesting that prompt engineering is crucial to improve zero-shot classification while different wording and phrasing can lead to inconsistencies. The effect of including definitions of fallacies had mixed outcomes; while it improved the performance of Meta-Llama-3.1-8B-Instruct on prompt 2, it slightly lowered its accuracy with prompt 1. For Mistral-7B-Instruct-v0.3, including definitions led to a decline in accuracy but increased the F1-score which means that it led to better classification for the less represented classes. On the other hand, Falcon-Mamba-7B-Instruct had consistently poor performance, and the inclusion of definitions seemed to increase the ratio of unknown labels. These findings show that while additional context may help certain models, it does so based on the capacity of the model to make use of the information given, and it may not always be beneficial.

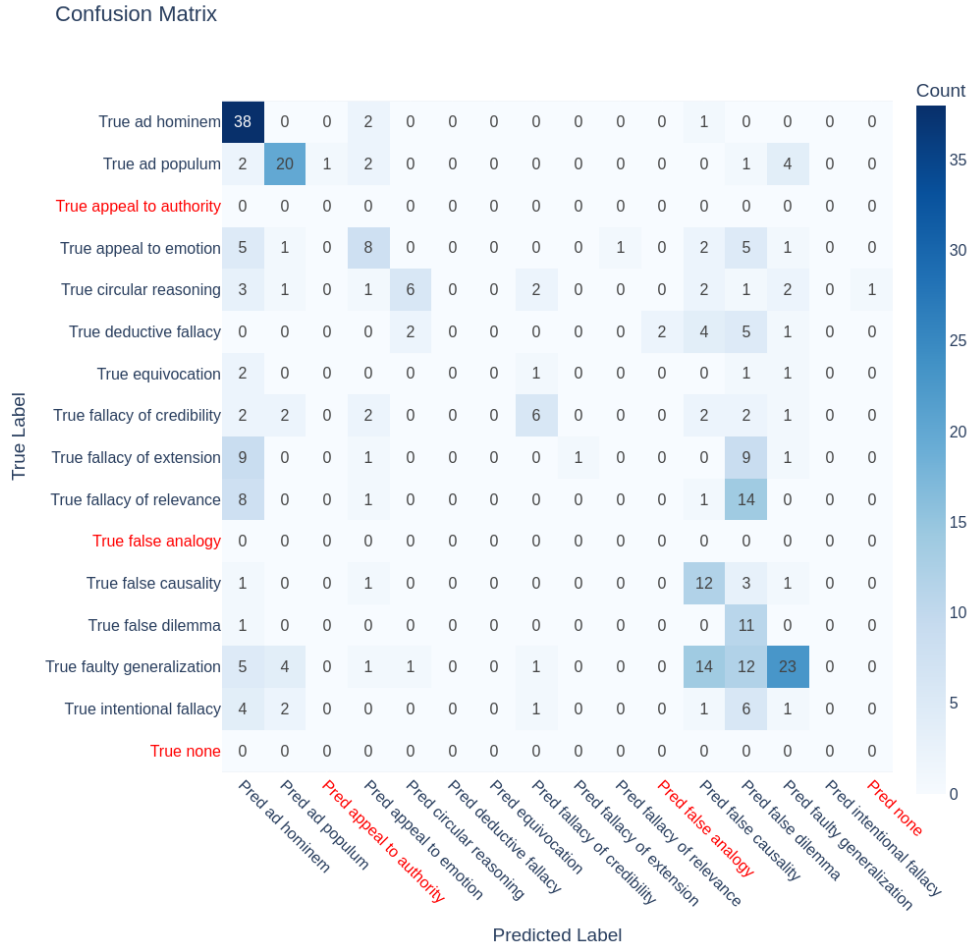


Figure 4.4: Best performing model’s results Confusion Matrix. The unknown labels (the ones that were not in the dataset) are highlighted with red.

To better understand the results presented, we provide the Confusion Matrix of the best performing model (Figure 4.4), Meta-Llama-3.1-8B-Instruct with prompt 2 and definitions included. The confusion matrix shows the distribution of the predicted labels compared to the true labels. The unknown labels are highlighted in red. The confusion matrix shows that the model had the most difficulty with the labels that are less represented in the dataset. Furthermore, the model had a hard time distinguishing between the labels that are similar to each other, such as *Faulty Generalization* and *False Dilemma* and *False Causality*.

4.2.2 Coarse Grained Copi’s Classes

In this section, we shift the classification task to Copi’s four coarse-grained categories: Fallacies of Relevance, Defective Induction, Presumption, and Ambiguity.

model	def	prompt	accuracy (%)	F1-score	failed (%)	unknown (%)
Random-Baseline	N/A	N/A	25.866	0.199	0.000	0.000
Distribution-Baseline	N/A	N/A	42.933	0.242	0.000	0.000
falcon-mamba-7b-instruct	False	1	43.667	0.080	2.000	2.667
falcon-mamba-7b-instruct	True	1	51.000	0.134	5.000	1.333
Meta-Llama-3.1-8B-Instruct	False	1	55.333	0.371	0.000	0.000
Meta-Llama-3.1-8B-Instruct	True	1	61.000	0.408	0.000	0.000
Mistral-7B-Instruct-v0.3	False	1	44.333	0.102	0.000	4.667
Mistral-7B-Instruct-v0.3	True	1	50.333	0.083	0.000	2.667
falcon-mamba-7b-instruct	False	2	30.333	0.036	0.333	9.333
falcon-mamba-7b-instruct	True	2	38.667	0.056	1.000	9.000
Meta-Llama-3.1-8B-Instruct	False	2	59.000	0.327	0.333	0.000
Meta-Llama-3.1-8B-Instruct	True	2	54.333	0.312	0.333	0.000
Mistral-7B-Instruct-v0.3	False	2	33.333	0.060	0.000	11.667
Mistral-7B-Instruct-v0.3	True	2	47.667	0.069	0.000	12.333

Table 4.2: Coarse-grained Copi Classes Results

As expected, because of the lower number of classes, the results presented in the table 4.2 show that the models performed better when classifying the fallacies into Copi’s coarse-grained classes compared to the fine-grained classes. The two baselines (random and distribution-based) provide reference points, with the distribution baseline achieving an accuracy of 42.93% and F1-score of 24.2%.

Overall, Meta-Llama-3.1-8B-Instruct stands out, peaking at 61.00% accuracy and 40.8% macro F1-score when given definitions and Prompt 1, outperforming not only the baselines but also the other two LLMs by a considerable margin. Another interesting finding is that the Llama model performed a lot better with prompt 1 compared to prompt 2, which is the opposite of what was observed with the fine-grained classes.

Falcon-Mamba-7b-Instruct shows more volatile behavior, sometimes dropping below the distribution baseline when using Prompt 2 (e.g., 30.33% accuracy). By contrast, it reaches 51.00% accuracy with Prompt 1 plus definitions, indicating that the inclusion of definitions can improve its performance—though not consistently across prompts. Mistral-7B-Instruct-v0.3 also benefits somewhat from definitions (e.g., rising from 33.33% to 47.67% accuracy with Prompt 2), yet exhibits very low F1-scores, lower than the baselines, suggesting difficulties in correctly classifying minority classes.



Figure 4.5: Best performing model’s results Confusion Matrix

The Confusion Matrix of the best performing model-Meta-Llama-3.1-8B-Instruct with prompt 1 and definitions included- is presented in Figure 4.5. Again, we notice that the model had the most difficulty with the labels that are less represented in the dataset. Furthermore, the model falsely classified 56 samples as *Fallacy of Relevance* which is the most common class in the dataset instead of the true label *Fallacy of Defective Induction*.

model	def	prompt	accuracy (%)	F1-score	failed (%)	unknown (%)
falcon-mamba-7b-instruct	False	1	54.000	0.276	10.000	0.000
falcon-mamba-7b-instruct	True	1	50.000	0.260	14.333	0.000
Meta-Llama-3.1-8B-Instruct	False	1	67.000	0.389	2.333	0.000
Meta-Llama-3.1-8B-Instruct	True	1	64.000	0.380	1.667	0.000
Mistral-7B-Instruct-v0.3	False	1	61.000	0.332	6.333	0.000
Mistral-7B-Instruct-v0.3	True	1	62.333	0.359	7.333	0.000
falcon-mamba-7b-instruct	False	2	52.333	0.280	15.333	0.000
falcon-mamba-7b-instruct	True	2	48.667	0.272	19.000	0.000
Meta-Llama-3.1-8B-Instruct	False	2	64.667	0.396	2.333	0.000
Meta-Llama-3.1-8B-Instruct	True	2	67.000	0.411	1.333	0.000
Mistral-7B-Instruct-v0.3	False	2	64.000	0.397	4.667	0.000
Mistral-7B-Instruct-v0.3	True	2	60.000	0.354	8.333	0.000

Table 4.3: Results when mapping fine-grained results to Copi’s coarse-grained classes

In Table 4.3, the model outputs from fine-grained classification are re-mapped to Copi’s four higher-level categories (Fallacies of Relevance, Defective Induction, Presumption, and Ambiguity). As a result, even if a model slightly misses the exact fine-grained label within one category (e.g., classifying on text segment as Ad Hominem whereas its true label is Appeal to Emotion), its coarse-grained mapping still can be correct. This characteristic explains why accuracy values here are notably higher than in direct fine-grained classification tasks. For instance, Meta-Llama-3.1-8B-Instruct attains 67% accuracy and a 41.1% F1 score with Prompt 2 and definitions included. Mistral-7B-Instruct-v0.3 similarly rises above 60% accuracy in several runs once its outputs are aggregated at the Copi category level. These results reinforce the notion that, while precise identification of individual fallacies is challenging, the models maintain enough conceptual understanding of logical fallacies to place them into the correct high-level group.

Overall, all three models here surpass the results typically seen in purely fine-grained tasks, implying that near-miss errors at the subtype level can still map into correct coarse-grained classes. This capacity to abstract from finely detailed labels to broader categories aligns well with the premise that the models possess inherent logical fallacy knowledge, even if they do not always apply it perfectly at the most specific level.

4.2.3 Coarse Grained Aristotle’s Classes

In this section, each model is evaluated on Aristotle’s three coarse-grained categories—Credibility (Ethos), Logic (Logos), and Emotion (Pathos)—alongside two baseline systems.

model	def	prompt	accuracy (%)	F1-score	failed (%)	unknown (%)
Random-Baseline	N/A	N/A	33.333	0.319	0.000	0.000
Distribution-Baseline	N/A	N/A	38.667	0.330	0.000	0.000
falcon-mamba-7b-instruct	False	1	36.333	0.127	1.333	1.667
falcon-mamba-7b-instruct	True	1	34.333	0.072	1.667	5.333
Meta-Llama-3.1-8B-Instruct	False	1	51.667	0.465	0.000	0.000
Meta-Llama-3.1-8B-Instruct	True	1	55.667	0.303	0.000	0.667
Mistral-7B-Instruct-v0.3	False	1	52.333	0.116	0.000	8.000
Mistral-7B-Instruct-v0.3	True	1	50.667	0.116	0.000	8.333
falcon-mamba-7b-instruct	False	2	41.333	0.136	0.333	2.333
falcon-mamba-7b-instruct	True	2	41.333	0.104	0.000	4.333
Meta-Llama-3.1-8B-Instruct	False	2	41.000	0.272	0.333	0.000
Meta-Llama-3.1-8B-Instruct	True	2	54.333	0.292	0.333	0.333
Mistral-7B-Instruct-v0.3	False	2	56.333	0.094	0.000	10.000
Mistral-7B-Instruct-v0.3	True	2	59.667	0.285	0.000	1.667

Table 4.4: Coarse-grained Aritotle Classes Results

In Table 4.4, comparing first to the random and distribution baselines, most model-prompt-definition configurations outperform these references in accuracy, although the degree of improvement differs substantially. In contrast all models - with the exception of Meta-Llama-3.1-8B-Instruct with prompt 1 and not including definitions - are outperformed by the baselines in F1-score suggesting that they struggle to correctly classify the minority classes.

In particular, Mistral-7B-Instruct-v0.3 when provided with definitions and using prompt 2 attains the highest overall accuracy at 59.67%, yet its macro F1 remains relatively modest (0.285), implying disproportionate success with certain classes and more confusion among others. Another interesting observation for Mistral-7B-Instruct-v0.3 is that it performs substantially better when provided with definitions in the prompt 2 setting where its accuracy rises from 56.33% to 59.67% and its F1-score from 0.094 to 0.285.

By contrast, Meta-Llama-3.1-8B-Instruct displays a more balanced performance pattern, most notably when using prompt 1 and no definitions are provided, yielding a 51.67% accuracy and an F1 of 0.465—the highest F1 of any configuration. This higher F1 indicates that the model better distributes predictions more evenly. Meanwhile, falcon-mamba-7b-instruct shows less consistent gains and struggles to surpass the baselines especially in F1-score where it is significantly outperformed by the baselines.

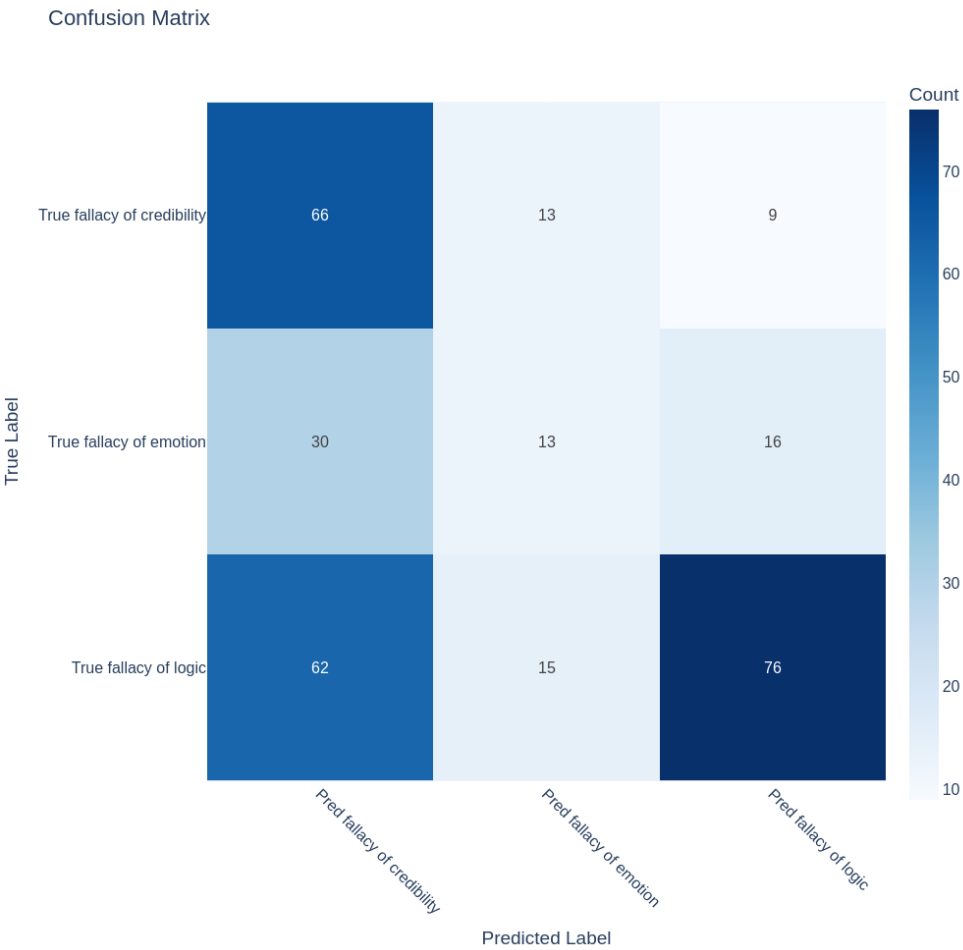


Figure 4.6: Best performing model’s results Confusion Matrix

In Figure 4.6, the confusion matrix of the best performing model in terms of macro F1-score (Meta-Llama-3.1-8B-Instruct with prompt 1 and no definitions) is presented. The confusion matrix shows that the model seriously overestimates the *Fallacy of Credibility* class with 62 samples classified as this class belonging to the *Fallacy of Logic* class and 30 samples belonging to the *Fallacy of Emotion* class.

model	def	prompt	accuracy (%)	F1-score	failed (%)	unknown (%)
falcon-mamba-7b-instruct	False	1	41.333	0.328	10.000	0.000
falcon-mamba-7b-instruct	True	1	39.000	0.303	14.333	0.000
Meta-Llama-3.1-8B-Instruct	False	1	60.333	0.422	2.333	0.000
Meta-Llama-3.1-8B-Instruct	True	1	60.333	0.435	1.667	0.000
Mistral-7B-Instruct-v0.3	False	1	57.667	0.456	6.333	0.000
Mistral-7B-Instruct-v0.3	True	1	53.667	0.422	7.333	0.000
falcon-mamba-7b-instruct	False	2	41.667	0.352	15.333	0.000
falcon-mamba-7b-instruct	True	2	40.000	0.329	19.000	0.000
Meta-Llama-3.1-8B-Instruct	False	2	63.333	0.446	2.333	0.000
Meta-Llama-3.1-8B-Instruct	True	2	62.333	0.465	1.333	0.000
Mistral-7B-Instruct-v0.3	False	2	59.333	0.402	4.667	0.000
Mistral-7B-Instruct-v0.3	True	2	52.333	0.360	8.333	0.000

Table 4.5: Results when mapping fine-grained results to Aristotle’s coarse-grained classes

In Table 4.5, each model’s fine-grained predictions are re-mapped into Aristotle’s three top-level categories (Credibility/Ethos, Logic/Logos, and Emotion/Pathos). Similar to the Copi-based aggregation, this strategy substantially boosts overall accuracy and reduces confusion compared to purely fine-grained classification. Notably, Meta-Llama-3.1-8B-Instruct stands out again, achieving up to 63.33% accuracy and 0.465 macro F1, underscoring its ability to preserve broad distinctions among ethos-, logos-, and pathos-based fallacies even if it occasionally mislabels more specific subtypes.

Overall, these results confirm that while fine-grained mislabeling can still occur, aggregating fallacies into Aristotle’s more general ethos–logos–pathos schema mitigates many classification mistakes, reflecting an underlying conceptual grasp of these categories.

4.3 Chain of Thought (CoT)

Having established the zero-shot performance of each model, we next explore whether guiding the models with basic chain-of-thought (CoT) prompts can improve their ability to detect logical fallacies. More specifically, we used the same setup as in the previous section with the only change of adding the following phrase to the prompt:

"Think step by step and evaluate all the possible fallacies listed above before responding!"

This section examines both fine-grained and coarse-grained classification under CoT conditions, analyzing how step-by-step reasoning influences accuracy, macro F1, and error rates for each model configuration compared to the results we obtained in the zero-shot setting.

4.3.1 Fine Grained Logic Classes

Looking at the Table 4.6 , several noteworthy trends emerge. In general, Meta-Llama-3.1-8B-Instruct continues to exhibit the strongest performance, marginally improving from 41.67% accuracy and 0.237 F1 from the zero-shot experiment to 42.00% accuracy and 0.266 F1 under Prompt 2 with definitions. This uptick is particularly significant in the F1 score, suggesting that CoT prompts help the model distribute more evenly its predictions across all classes. By contrast, Mistral-7B-Instruct-v0.3 sees a slight decline or inconsistent changes in accuracy and F1 when chain-of-thought is employed, indicating that the extra instructions do not always translate into clearer distinctions between closely related fallacies.

Falcon-Mamba-7B-Instruct generally struggles under both setups, with chain-of-thought sometimes reducing accuracy (e.g., from 24.67% to 22.67% in Prompt 2, no definitions). It also exhibits elevated “failed” or “unknown” percentages, suggesting that basic CoT prompts alone are insufficient to mitigate formatting errors or prevent it from generating out-of-set labels. Therefore, while short chain-of-thought rationales do benefit the stronger model (Meta-Llama-3.1-8B-Instruct), the results underscore that the approach yields at best modest improvements, and its effectiveness varies considerably across different LLM architectures.

model	def	prompt	accuracy (%)	F1-score	failed (%)	unknown (%)
falcon-mamba-7b-instruct	False	1	17.667	0.062	5.000	9.000
falcon-mamba-7b-instruct	True	1	18.333	0.081	0.333	5.333
Meta-Llama-3.1-8B-Instruct	False	1	39.333	0.213	0.000	3.000
Meta-Llama-3.1-8B-Instruct	True	1	36.667	0.251	0.000	0.333
Mistral-7B-Instruct-v0.3	False	1	39.667	0.161	0.000	5.333
Mistral-7B-Instruct-v0.3	True	1	34.333	0.198	0.000	7.333
falcon-mamba-7b-instruct	False	2	22.667	0.087	2.333	10.000
falcon-mamba-7b-instruct	True	2	23.000	0.081	3.000	15.000
Meta-Llama-3.1-8B-Instruct	False	2	41.000	0.230	0.000	3.000
Meta-Llama-3.1-8B-Instruct	True	2	42.000	0.266	0.000	0.667
Mistral-7B-Instruct-v0.3	False	2	37.667	0.171	0.000	5.000
Mistral-7B-Instruct-v0.3	True	2	32.333	0.162	0.000	7.333

Table 4.6: Fine-grained Classes Results with CoT prompting

4.3.2 Coarse Grained Copi’s Classes

Comparing the Table 4.7 results with the zero-shot experiment’s results (Table 4.2) reveals interesting shifts in each model’s ability to classify fallacies under Copi’s four broad categories. Meta-Llama-3.1-8B-Instruct generally reaffirms its leading position, but the magnitude of improvement varies by prompt and whether definitions are provided. For instance, when no definitions are provided using Prompt 1, Meta-Llama-3.1-8B-Instruct rises from 55.33% accuracy and 0.371 F1 to 59.67% accuracy and 0.405 F1 —a solid gain in both metrics. Meanwhile, its performance with definitions under

the same prompt remains stable (61.00% accuracy with or without CoT), suggesting diminishing returns if the model is already near its high-end performance.

The other two models see more uneven effects from CoT prompting. Falcon-Mamba-7B-Instruct, for example, experiences a slight jump under some settings (e.g., with definitions using Prompt 1 from 51.00% to 53.33% accuracy), but falls behind under Prompt 2, where both accuracy and F1 can drop well below those of the baseline. Similarly, Mistral-7B-Instruct-v0.3 occasionally benefits—moving from 38.67% to 50.67% accuracy under Prompt 1 with definitions—but also suffers decreased performance in Prompt 2 scenarios, as indicated by high unknown label rates (up to 13.33%) and minimal F1 gains. Overall, chain-of-thought prompts appear to help Meta-Llama-3.1-8B-Instruct refine its category decisions, whereas Falcon-Mamba-7B-Instruct and Mistral-7B-Instruct-v0.3 exhibit more variability, underscoring that the effectiveness of step-by-step reasoning largely depends on a model’s baseline strengths and its sensitivity to prompt style.

model	def	prompt	accuracy (%)	F1-score	failed (%)	unknown (%)
falcon-mamba-7b-instruct	False	1	49.333	0.102	2.667	3.000
falcon-mamba-7b-instruct	True	1	53.333	0.132	1.000	0.667
Meta-Llama-3.1-8B-Instruct	False	1	59.667	0.405	0.000	0.000
Meta-Llama-3.1-8B-Instruct	True	1	61.000	0.403	0.000	0.000
Mistral-7B-Instruct-v0.3	False	1	38.667	0.089	0.000	3.667
Mistral-7B-Instruct-v0.3	True	1	50.667	0.102	0.000	1.667
falcon-mamba-7b-instruct	False	2	34.667	0.071	2.667	5.667
falcon-mamba-7b-instruct	True	2	38.333	0.070	0.000	12.333
Meta-Llama-3.1-8B-Instruct	False	2	58.000	0.388	0.000	0.000
Meta-Llama-3.1-8B-Instruct	True	2	51.333	0.284	0.333	0.000
Mistral-7B-Instruct-v0.3	False	2	30.667	0.051	0.000	12.000
Mistral-7B-Instruct-v0.3	True	2	43.667	0.064	0.000	13.333

Table 4.7: Coarse-grained Copi’s Classes Results with CoT prompting

4.3.3 Coarse Grained Aristotle’s Classes

In Table 4.8, Meta-Llama-3.1-8B-Instruct generally maintains or slightly improves its strong performance compared to the zero-shot results (Table 4.4). For instance, under Prompt 1 and with definitions, its accuracy increases from 55.67% to 58.00%, while the macro F1 also rises substantially (0.303 to 0.379). By contrast, Falcon-Mamba-7B-Instruct sees more negative effects of CoT in the same setting, with its accuracy dropping noticeably (e.g., from 36.33% to 27.33% when no definitions are provided) or remaining relatively low. Mistral-7B-Instruct-v0.3 also experiences mixed results: while accuracy sometimes moves upward with CoT (e.g., from 52.33% to 55.67% when using prompt 1 and no definitions are provided), the macro F1 remains modest, reflecting ongoing difficulty differentiating less frequent classes under Aristotle’s three broad categories (Credibility/Ethos, Logic/Logos, Emotion/Pathos).

Additional textual definitions and variations in prompt style further influence these outcomes. Meta-Llama-3.1-8B-Instruct benefits more consistently from defini-

tions, especially in Prompt 1, achieving stable or slightly higher accuracy without harming its F1 score. Mistral-7B-Instruct-v0.3, by contrast, can handle definitions better under some conditions (e.g., Prompt 2) but still struggles to maintain balanced classification, as indicated by low F1 or higher unknown label rates. Falcon-Mamba-7B-Instruct remains the most susceptible to confusion: although the chain-of-thought framework can sometimes resolve trivial mislabels, it often triggers formatting errors or leads to higher “unknown” outputs. Consequently, while CoT prompts bring incremental improvements for the best-performing model, they do not reliably elevate the other two models to a better performance in Aristotle-based fallacy classification.

model	def	prompt	accuracy (%)	F1-score	failed (%)	unknown (%)
falcon-mamba-7b-instruct	False	1	27.333	0.078	3.333	3.333
falcon-mamba-7b-instruct	True	1	29.667	0.079	1.000	2.333
Meta-Llama-3.1-8B-Instruct	False	1	52.667	0.437	0.000	0.000
Meta-Llama-3.1-8B-Instruct	True	1	58.000	0.379	0.333	0.000
Mistral-7B-Instruct-v0.3	False	1	55.667	0.158	0.000	4.333
Mistral-7B-Instruct-v0.3	True	1	55.000	0.175	0.000	4.667
falcon-mamba-7b-instruct	False	2	38.000	0.137	1.000	1.333
falcon-mamba-7b-instruct	True	2	39.333	0.085	0.667	3.333
Meta-Llama-3.1-8B-Instruct	False	2	39.000	0.253	0.333	0.000
Meta-Llama-3.1-8B-Instruct	True	2	53.667	0.377	0.333	0.000
Mistral-7B-Instruct-v0.3	False	2	54.000	0.091	0.000	10.333
Mistral-7B-Instruct-v0.3	True	2	59.000	0.190	0.000	3.000

Table 4.8: Coarse-grained Aristotle’s Classes Results with CoT prompting

Altogether, our results in the basic CoT setting indicate that while step-by-step prompts offer modest gains. Models like Meta-Llama-3.1-8B-Instruct benefit the most, seeing slight upticks in both accuracy and F1, whereas weaker baselines and prompts remain prone to misclassification or labeling errors.

4.4 Multi-Round CoT (From Coarse-Grained to Fine-Grained)

In this section we explore the impact of multi-round Chain-of-Thought (CoT) reasoning on each model’s ability to classify logical fallacies. We first present the results when going from Copi’s coarse-grained categories to fine-grained classes and then from Aristotle’s coarse-grained categories to fine-grained classes. The goal is to assess whether multi-round reasoning can help models refine their predictions and reduce misclassification errors compared to the zero-shot setup.

4.4.1 Coarse Grained Copi’s Classes to fine-grained classes

In comparing the multi-round Chain-of-Thought results shown in Table 4.9 with the zero-shot fine-grained outcomes in Table 4.1, we observe that the multi-round approach does not always lead to higher final accuracy or macro F1. Falcon-Mamba-7B-Instruct, for instance, exhibits drastically lower final accuracy (often below 10%) in the multi-round setting compared to its zero-shot results (roughly 20–24%). By having a closer look to its responses we realized that it actually failed in most cases to change its response from the first round. Even Meta-Llama-3.1-8B-Instruct and Mistral-7B-Instruct-v0.3, which perform substantially better than Falcon-Mamba, generally see a dip in accuracy under multi-round classification; Meta-Llama’s best multi-round accuracy (32.67%) remains well below its strongest zero-shot accuracy (41.67%). Nevertheless, in this setting its best F1-score is a bit higher than the one in the zero-shot experiment, suggesting that this method helped it create more balanced predictions across all classes. However, the added complexity of having to correctly classify the top-level category first which means that any coarse-level error propagates to the final label, effectively magnified the overall misclassification rate. Overall, the results underscore that, while a hierarchical classification scheme can offer a boost in the classification of minority classes, it introduces additional points of failure that can outweigh potential benefits.

model	def	prompt	acc (%)	F1-score	failed (%)	unknown (%)
falcon-mamba-7b-instruct	False	1	5.000	0.032	0.333	60.000
falcon-mamba-7b-instruct	True	1	7.000	0.025	0.333	53.000
Meta-Llama-3.1-8B-Instruct	False	1	32.667	0.213	0.000	1.333
Meta-Llama-3.1-8B-Instruct	True	1	31.667	0.212	0.000	1.333
Mistral-7B-Instruct-v0.3	False	1	30.333	0.179	0.000	1.333
Mistral-7B-Instruct-v0.3	True	1	28.667	0.176	0.000	1.333
falcon-mamba-7b-instruct	False	2	8.333	0.045	0.000	50.333
falcon-mamba-7b-instruct	True	2	11.333	0.051	0.000	29.333
Meta-Llama-3.1-8B-Instruct	False	2	32.000	0.231	0.000	0.333
Meta-Llama-3.1-8B-Instruct	True	2	32.667	0.247	0.000	0.667
Mistral-7B-Instruct-v0.3	False	2	31.000	0.157	0.000	1.667
Mistral-7B-Instruct-v0.3	True	2	34.333	0.160	0.000	4.333

Table 4.9: Copi’s Coarse-Grained to Fine-Grained Multi-round CoT

4.4.2 Coarse Grained Aristotle’s Classes to fine-grained classes

In Table 4.10, the multi-round Chain-of-Thought (CoT) approach for Aristotle’s three top-level categories (Credibility, Logic, Emotion) is presented. Compared to the zero-shot fine-grained results, it generally yields lower final accuracy and macro F1-score across most configurations. For instance, Meta-Llama-3.1-8B-Instruct achieved more than 40% accuracy in its strongest zero-shot fine-grained setting (see Table 4.1), whereas here it peaks at about 35% accuracy. In contrast, the F1-score peaked at 0.247

in this case, while the maximum in the zero shot setting was 0.237. Falcon-Mamba-7B-Instruct, as in the previous Table 4.9 drops below 12% accuracy, and even Mistral-7B-Instruct-v0.3’s best performance in these runs (roughly 35.7% accuracy) remains lower than the upper 30s or low 40s it could reach in direct fine-grained classification. These results, in general echoe the observations from Copi’s multi-round scheme, indicating that requiring correct coarse-level decisions first often creates more failure points in low-resource or zero-shot scenarios.

model	def	prompt	acc (%)	F1-score	failed (%)	unknown (%)
falcon-mamba-7b-instruct	False	1	7.333	0.035	0.333	31.000
falcon-mamba-7b-instruct	True	1	10.333	0.054	0.333	20.667
Meta-Llama-3.1-8B-Instruct	False	1	32.333	0.215	0.000	1.000
Meta-Llama-3.1-8B-Instruct	True	1	32.000	0.233	0.000	3.333
Mistral-7B-Instruct-v0.3	False	1	35.333	0.128	0.000	5.000
Mistral-7B-Instruct-v0.3	True	1	26.667	0.134	0.000	7.000
falcon-mamba-7b-instruct	False	2	7.667	0.043	0.000	39.333
falcon-mamba-7b-instruct	True	2	11.667	0.069	0.000	18.000
Meta-Llama-3.1-8B-Instruct	False	2	30.333	0.196	0.000	1.000
Meta-Llama-3.1-8B-Instruct	True	2	35.333	0.250	0.000	4.333
Mistral-7B-Instruct-v0.3	False	2	35.667	0.188	0.000	3.000
Mistral-7B-Instruct-v0.3	True	2	31.333	0.175	0.000	11.333

Table 4.10: Aristotle’s Coarse-Grained to Fine-Grained Multi-round CoT

Overall, the multi-round CoT experiments didn’t lead to increased performance. F1-score increased a little in some cases whereas accuracy dropped significantly in most cases. The falcon-mamba failed completely on this approach, not being able to differentiate its second response in a lot of cases from the first one. The fact that any missteps in the initial coarse-grained stage propagate to the second round, frequently led to worse performance compared to single-step, fine-grained classification. This trend holds for both Copi’s four-category scheme and Aristotle’s three-category approach, although the severity of performance drops varies by model and prompt format. Consequently, while multi-round CoT offers an appealing logical framework for detecting logical fallacies, the results suggest that this method does not substantially help the models to distinguish the different fine-grained classes, even when their top-level classification is correct.

4.5 Semantically Unrelated Labels

In this additional experiment, we aimed to assess the extent to which LLMs rely on the semantics of predefined fallacy labels versus their ability to infer meaning from contextual definitions. Each model was presented with a list of semantically unrelated fallacy classes, labeled generically as *Fallacy_1*, *Fallacy_2*, ..., accompanied by the

corresponding definitions of their actual mapped fallacy categories. The results are presented in the following tables and are also compared with the zero-shot results.

4.5.1 Fine Grained Logic Classes

When presented with semantically unrelated label names, the models exhibit intriguing shifts in both accuracy and macro F1 compared to the Zero-Shot experiments. As shown in Table 4.11 compared to Table 4.1, Meta-Llama-3.1-8B-Instruct experiences a modest drop in accuracy (from roughly 41.6% to 39.3%), yet demonstrates a notable rise in F1-score (from 0.237 to 0.287), indicating that it more effectively balances predictions across classes under these altered label conditions. Mistral-7B-Instruct-v0.3 displays a similar pattern, with a slight reduction in accuracy offset by an increased F1-score, further suggesting improved coverage of low-frequency fallacies. By contrast, Falcon-Mamba-7B-Instruct performs poorly under this setup, underscoring that it relies heavily on the semantic cues of known label names learned during training. Another significant observation is the absence of 'failed' and 'unknown' labels, particularly in weaker models such as Falcon-Mamba. Taken together, these findings imply that while the strongest models can adapt to different naming conventions and potentially enhance minority-class performance, weaker architectures may depend more on memorized lexical mappings and struggle without familiar label semantics.

model	prompt	accuracy (%)	F1-score	failed (%)	unknown (%)
falcon-mamba-7b-instruct	1	9.333	0.035	0.667	0.667
Meta-Llama-3.1-8B-Instruct	1	35.667	0.253	0.000	0.000
Mistral-7B-Instruct-v0.3	1	34.000	0.261	0.000	0.000
falcon-mamba-7b-instruct	2	10.333	0.050	1.000	0.333
Meta-Llama-3.1-8B-Instruct	2	39.333	0.287	0.333	0.000
Mistral-7B-Instruct-v0.3	2	34.667	0.248	0.000	0.000

Table 4.11: Fine-Grained Classes Results with Semantically Unrelated Labels

4.5.2 Copi's Coarse Grained Classes

In Table 4.12, the trends are different when the models are presented with semantically unrelated labels in the Copi's four categories. Meta-Llama-3.1-8B-Instruct and Mistral-7B-Instruct-v0.3 both exhibit a decrease in accuracy and F1-score compared to the Zero-Shot setting when definitions were included. The 'failed' and 'unknown' percentages have vanished in this case too.

model	prompt	accuracy (%)	F1-score	failed (%)	unknown (%)
falcon-mamba-7b-instruct	1	35.667	0.063	0.667	1.667
Meta-Llama-3.1-8B-Instruct	1	38.000	0.225	0.000	0.000
Mistral-7B-Instruct-v0.3	1	46.667	0.214	0.000	0.000
falcon-mamba-7b-instruct	2	42.667	0.073	0.667	1.667
Meta-Llama-3.1-8B-Instruct	2	34.333	0.164	0.667	0.000
Mistral-7B-Instruct-v0.3	2	31.667	0.201	0.000	0.000

Table 4.12: Copi’s Classes Results with Semantically Unrelated Labels

4.5.3 Aristotle’s Coarse Grained Classes

Finally, in Table 4.13, the trends are more similar with the results for fine-grained classes rather than the Copi’s coarse grained categories. Meta-Llama-3.1-8B-Instruct and Mistral-7B-Instruct-v0.3 both exhibit a slight decrease in accuracy compared to the Zero-Shot setting when definitions were included, whereas the F1-score faces a significant increase in some cases. Falcon-Mamba-7B-Instruct, once again struggles to achieve a decent performance in this setting, even though ‘failed’ and ‘unknown’ percentages which was one of the issues that this model had in the previous experiments, are significantly lower here.

model	prompt	accuracy	F1-score	failed (%)	unknown (%)
falcon-mamba-7b-instruct	1	20.667	0.049	0.333	3.000
Meta-Llama-3.1-8B-Instruct	1	51.000	0.463	0.000	0.000
Mistral-7B-Instruct-v0.3	1	48.000	0.351	0.000	0.333
falcon-mamba-7b-instruct	2	32.333	0.136	1.333	1.000
Meta-Llama-3.1-8B-Instruct	2	43.667	0.333	1.000	0.000
Mistral-7B-Instruct-v0.3	2	46.667	0.278	0.000	0.667

Table 4.13: Aristotle’s Classes Results with Semantically Unrelated Labels

These findings highlight that models like Meta-Llama-3.1-8B-Instruct and Mistral-7B-Instruct-v0.3 demonstrate resilience by maintaining reasonable classification performance despite semantically unrelated labels, and are even able to overcome biases from their training data by performing better in this setting in minority classes. On the other hand, weaker models such as Falcon-Mamba-7B-Instruct struggle significantly, relying heavily on explicit lexical cues.

CHAPTER 5

Conclusion

This thesis examined whether large language models, used in a zero-shot capacity, could effectively identify logical fallacies across a range of categories, a task that has been found to be difficult even for human annotators who are capable of meaningfully perceiving fallacies in different ways. Drawing on the LOGIC dataset, our experiments examined both fine-grained and coarse-grained fallacy classifications through various prompting strategies, including explicit definitions, chain-of-thought (CoT) methods, and multi-round hierarchical approaches. The results underscore several key observations.

Regarding Zero-Shot Viability, even without domain-specific fine-tuning, models such as Meta-Llama-3.1-8B-Instruct and Mistral-7B-Instruct-v0.3 show non-trivial capabilities in fallacy recognition. Nevertheless, accuracy and macro F1 scores remain far from perfect, indicating a persistent gap between zero-shot classification and human-level reasoning. Furthermore, the explicit inclusion of definitions also offered mixed results, suggesting that while additional textual cues can sometimes enhance classification, they can also introduce noise or confusion. Additionally, the results suggest - from mapping the model's fine-grained predictions to Copi's and Aristotle's broader categories - that the models possess inherent logical fallacy understanding, even if they do not always apply it perfectly at the most specific level.

On Impact of Prompt Design, prompt structure—particularly the choice between two slightly different prompts—proved to be a critical factor, often shifting accuracy and F1 by notable margins. This sensitivity underscores the importance of prompt engineering in zero-shot classification settings.

For Hierarchical Classification, Multi-round CoT schemes, in which a model predicts a coarse category before refining to a specific subtype, tended to introduce an extra point of failure. If the model misclassified the top-level category, its final prediction was certainly wrong. As a result, overall performance did not consistently surpass that of single-step zero-shot classification but did offer some gains in F1-score especially in the strongest models.

Finally, familiar labels (e.g., “Ad Hominem”) were replaced with neutral placeholders (e.g., “Fallacy_1”). Interestingly, Meta-Llama and Mistral sometimes preserved similar or even improved macro F1, indicating they could compensate for lost semantic clues by relying on contextual cues in the definitions, and thus moving away from any bias from their training data. Falcon-Mamba, however, suffered significant performance declines, highlighting its heavier dependence on memorized label semantics

rather than in-context learning.

In conclusion, our work contributes in understanding not only how large language models might assist in logical fallacy detection, but also what their current limitations are. As misinformation and flawed reasoning persist in public discourse, the continued evolution of fallacy detection methods could play a crucial role in fostering more critical, reasoned dialogue. By refining prompt engineering, exploring more sophisticated reasoning strategies, and integrating external knowledge sources, we can move closer to automated systems that not only detect fallacies but also provide transparent justifications for their decisions.

CHAPTER 6

Future Work

Moving forward, several directions could help refine and extend the work presented in this thesis.

First, an appropriate path for further research would be to try out more recent iterations of the models employed in this work. Large language model development is moving very fast, and each new iteration improves on reasoning capacity and understanding of context. Trying out current models like newer iterations of Meta-Llama, Falcon-Mamba, and Mistral might give more accurate results and give us a contemporary report on state-of-the-art fallacy detection capabilities.

Also, future research would investigate how model size affects performance measures. This research compared models of roughly equal parameter sizes; however, a comparison of a wider range of models—including smaller as well as larger and more powerful models—would enable a better understanding of the trade-offs that can be made between computational efficiency and classification accuracy. This kind of analysis would also inform us whether smaller models, can achieve the performance of their larger versions and consequently improve accessibility for applications with limited resources.

Another path could be adding justification generation to the fallacy detection process. Detecting a fallacy is often less meaningful if users cannot understand the model's reasoning. Future work might couple classification outputs with short, automatically generated justifications that illuminate why a particular argument is fallacious, thereby supporting both interpretability and educational objectives.

Finally, the inclusion of external knowledge sources, would also enhance logical fallacy detection. Hybrid approaches leveraging LLMs and retrieval-augmented generation (RAG) can further enhance the ability of LLMs in fallacy-detection tasks.

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APPENDIX A

Logical Fallacy Definitions

This appendix provides an overview of the logical fallacies from the LOGIC dataset [19] and their definitions used in the present work. Furthermore, we provide Copi's and Aristotle's broader categories alongside their definitions.

A.1 LOGIC Dataset Fallacies and their definitions

1 Appeal To Emotion: Attempting to arouse non-rational sentiments within
2 the intended audience in order to persuade.

3 False Causality: Occurs when someone mistakenly assumes that because one
4 event follows another, the first event caused the second, without
5 sufficient evidence for a causal link.

6 Ad Hominem: The opponent attacks a person instead of arguing against the
7 claims that the person has put forward.

8 Faulty Generalization: A generalization is drawn from a sample which is
9 too small, it is not representative of the population or it is not
10 applicable to the situation if all the variables are taken into account.

11 False Dilemma: Presenting two alternative options as the only
12 possibilities, when in fact more possibilities exist. As an extreme case
13 , telling the audience exactly what actions to take, eliminating any
14 other possible choices.

15 Fallacy of Relevance: The argument supporting the claim diverts
16 attention to issues that are irrelevant to the claim at hand.

17 Fallacy of Credibility: Happens when someone argues that a claim is true
18 simply because an authority or expert believes it, even if that
19 authority is not a reliable or relevant source on the topic.

20 Ad Populum: A fallacious argument based on affirming that something is
21 real or better because the majority thinks so.

17	Circular Reasoning: A fallacy where the end of an argument comes back to the beginning without having proven itself.
18	
19	Fallacy of Extension: An argument that attacks an exaggerated or caricatured version of the 'opponents position.
20	
21	Equivocation: An argument that uses a key term or phrase in an ambiguous way, with one meaning in one portion of the argument and another meaning in another portion.
22	
23	Deductive Fallacy: An error in the logical structure of an argument.
24	
25	Intentional Fallacy: Some intentional (sometimes subconscious) action/choice to incorrectly support an argument.
26	

Fallacy Definitions A.1: List of Logical Fallacies and Their Definitions

A.2 Copi's Fallacy Categories and their definitions

1	Fallacy of Relevance: These occur when the premises are not logically relevant to the conclusion, often relying on emotional appeals, distractions, or personal attacks rather than valid reasoning.
2	
3	Fallacy of Defective Induction: These involve weak or insufficient evidence leading to a conclusion, where the premises fail to provide strong support for the argument.
4	
5	Fallacy of Ambiguity: These arise from unclear or misleading language that creates confusion or misinterpretation, often through ambiguous wording or structure.
6	
7	Fallacy of Presumption: These occur when an argument assumes something without proper justification, relying on unstated or unsupported premises that fail to validate the conclusion.

Fallacy Definitions A.2: List of Copi's Coarse-Grained Fallacies and Their Definitions

A.3 Aristotle's Fallacy Categories and their definitions

1	Fallacy of Credibility: These fallacies occur when the appeal to credibility or character is flawed or misused.
2	
3	Fallacy of Logic: These fallacies occur when the logical structure of an argument is flawed.
4	

5 | `Fallacy of Emotion: These fallacies involve manipulative emotional appeals
that bypass logical reasoning.`

Fallacy Definitions A.3: List of Aristotle's Coarse-Grained Fallacies and Their Definitions

A.4 Source Code

Finally, the source code used for the present work can be found in this link:
<https://github.com/lefteriskat/fallacy-detection>.

